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Inventor(s)	Bouma; Katarina et al.

Methods and Apparatus for Using a Pallet Position Sensor Output To Facilitate Safe Material Handling Operations

Abstract

A pallet position sensor is used in some embodiments to determine when nesting of a pallet against a fork backrest has been achieved. If nesting has not been achieved following a pallet pickup operation a double bite procedure is automatically implemented in some embodiments in response to the pallet position sensor failing to detect successful pallet nesting on the forks. The use of the pallet position sensor allows for reliable confirmation of nesting of the pallet regardless of whether a normal or double bite pick operation is implemented. By using the output of the sensor to confirm nesting and to trigger a double bite procedure when nesting is not achieved with a single fork placement, e.g., extension operation, safety and reliability is enhanced as compared to systems which do not include a distance sensor capable of sensing the pallet position at the front of the forklift assembly.

Inventors: **Bouma; Katarina (Sacramento, CA), Mason; Julian (Redwood City, CA), Schenck; Connor (Erie, CO)**

Applicant: **Third Wave Automation, Inc.** (Union City, CA)

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Background/Summary

RELATED APPLICATIONS [0001] The present application claims the benefit of U.S. Provisional Patent Application 63/563,360 which was filed Mar. 9, 2023 and which is hereby expressly incorporated by reference in its entirety.

FIELD

[0002] The present application relates to robotic devices, e.g., autonomous material handling vehicles such as forklifts, and, more particularly, to methods and apparatus for determining pallet position on a set of forks and using the determined pallet position in automatically controlling material handling operations.

BACKGROUND

[0003] A pallet's position on a set of forks can have a big impact on whether a load will shift or slide off the forks while a load is being transported from one location to another. In addition, having a load such as a pallet far out on the forks rather than up against the fork backrest can increase the chance of the forklift tipping over since the load is further out from the center of gravity of the forklift then would be the case is the load was positioned up against the fork backrest.

[0004] In most forklift operations it is up to a human operator to determine that the load being transported is properly positioned on the forks. The operator's ability to confirm pallet placement can be limited by obstructions, including a forklift mast, that might interfere with the operator's view this is true whether the operator is physically present at the forklift, e.g., operating as a forklift driver, or is a remote operator trying to control the forklift from a remote location through the use of a camera.

[0005] From a safety and efficiency perspective it would be desirable if methods and/or apparatus could be developed which would reduce the need for an operator to determine proper load placement and/or which would allow operations to reposition a load on a set of forks to be automatically initiated when a forklift lift operation fails to achieve suitable pallet placement on the forks.

SUMMARY

[0006] A pallet position sensor, e.g., a non-contact distance sensor, is used in some embodiments to determine when nesting of a pallet against a fork backrest has been achieved. If nesting has not been achieved following a pallet pickup operation, a double bite procedure is automatically implemented in some embodiments in response to the pallet position sensor failing to detect successful pallet nesting on the forks. In a double bite procedure, the pallet is moved in the direction of the forklift after being lifted, e.g., by retracting the extendable forks. The pallet is then lowered to the ground or surface from which it was lifted, e.g., pallet rack shelf, and the forks are once again extended until nesting is achieved. Once nesting is achieved the pallet is raised and the forklift moves to the intended destination.

[0007] A double bite procedure is also useful when an initial attempt to move a pallet fails to achieve pallet nesting or when a pallet can not be reached and positioned into the nesting position in a single lift operation, e.g., because of an obstruction such as a raised curb, pallet rack support or overhang, that prevents the material handling device, e.g., robotic forklift from being positioned close enough to a pallet to be moved to achieve nesting in a single fork extension operation. In such a case the forklift can be positioned as close to the pallet as reasonably possible. A lift operation implemented with the pallet not being fully nested can be dangerous since failure to have the pallet fully nested increases the risk of pallet movement during transit and/or because having a load further out on the forks decreases forklift stability as compared to when the load if positioned fully on the forks in the nested position.

[0008] The use of the position sensor allows for reliable confirmation of nesting of the pallet regardless of whether a normal or double bite pick operation is implemented. By using the output of the sensor to confirm nesting and to trigger a double bite procedure when nesting is not achieved with a single fork placement, e.g., extension operation, safety and reliability is enhanced as compared to systems which do not include a distance sensor capable of sensing the pallet position at the front of the forklift assembly.

[0009] While various features and embodiments have been discussed in the summary above, it should be appreciated that not necessarily all embodiments include the same features and some of the features described above are not necessary but can be desirable in some embodiments. Numerous additional features, embodiments and benefits of various embodiments are discussed in the detailed description which follows.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0010] FIG. **1** is a drawing of a system implemented in accordance with one exemplary embodiment.

[0011] FIG. **2** illustrates an exemplary operator workstation that can be used as any of the operator workstations shown in FIG. **1**.

[0012] FIG. **3** illustrates an exemplary robotic device, e.g., a forklift which can operate autonomously, signals obstacle detection through the use of lights mounted on the robotic device, and in some embodiments includes a sensor for detecting when a pallet is properly nested on the forks of the robotic device.

[0013] FIG. **4** is a block diagram of the exemplary device of FIG. **3**, which includes an obstacle detection indicator light bar and pallet position sensor.

[0014] FIG. **5** is a flow chart showing the steps of an exemplary method implemented by an exemplary robotic device, such as the one shown in FIG. **3** or of any one of the other figures, used to detect obstacles and visually indicate obstacle detection information through the use of lights.

[0015] FIG. **6** shows an exemplary robotic device including obstruction detection indicator lights along a canopy and which includes a pallet position sensor in addition to a forward looking camera array.

[0016] FIG. **7** is a view of the robotic device of FIG. **6** from which the rear of the robotic device can be seen with the obstruction detection indicator lights along the rear of the canopy being visible.

[0017] FIG. **8** illustrates an exemplary robotic device in which an obstruction detection indicator light array is positioned along the top of the main body of the robotic device.

[0018] FIG. **9** is a view of the robotic device of FIG. **8** from which the rear of the robotic device can be seen with the obstruction detection indicator lights along the rear of the main body and rear door are visible.

[0019] FIG. **10** is a flow chart showing the steps of an exemplary method for controlling a robotic device to perform a pick operation, e.g., a pallet pick up operation.

[0020] FIG. **11** shows an example robotic device, e.g., robotic forklift, and fork distance associated with the robotic forklift.

[0021] FIG. **12** shows the forks of the robotic device of FIG. **11** in an extended position and the extension distance E that can be achieved by extending the forks in the exemplary embodiment.

[0022] FIG. **13** shows the robotic forklift of FIG. **11** on level ground in position to initiate a pallet pick up operation.
[0023] FIG. **14** shows the robotic forklift of FIG. **11** on level ground with the forks extended as part of a pallet pick up operation with the pallet being nested in the forks and ready for lifting from the ground.
[0024] FIG. **15** shows the robotic forklift of FIG. **11** on level ground with the forks extended as part of a pallet pick up operation with the pallet being nested in the forks and ready for lifting from the ground.
[0025] FIG. **16** shows the robotic forklift of FIG. **11** on ground that includes a raised area on which a pallet to be picked up is positioned.
[0026] FIG. **17** shows the robotic forklift of FIG. **11** performing the first step of a two bite (e.g., double bite) pickup procedure in which the pallet will be raised, moved, lowered and the forks reinserted to achieve proper pallet nesting before the robotic forklift moves to take the pallet to a new location.
[0027] FIG. **18** shows the robotic forklift of FIG. **11** lifting the pallet shown in FIG. **17** as part of the double bite pickup procedure.
[0028] FIG. **19** shows the robotic forklift of FIG. **11** retracting the pallet fork to move the pallet while it is in the air so it can be lowered and repositioned as part of the double bite pickup procedure.
[0029] FIG. **20** shows the robotic forklift of FIG. **11** after it has lowered the pallet to the ground as part of the double bite pickup procedure.
[0030] FIG. **21** shows the forks of the robotic forklift device of FIG. **11** reinserted into the pallet to achieve full nesting of the pallet as part of the double bite pickup procedure.
[0031] FIG. **22** shows the pallet having been lifted following full nesting of the pallet as part of the double bite pickup procedure and now being ready for transport.
[0032] FIG. **23** shows the pallet fully nested and being transported by the robotic forklift to a new location away from the raised shelf where the pallet was previously stored.
[0033] FIG. **24**A is a diagram showing a first part of a flow chart showing a method implemented in accordance with one embodiment of the invention.
[0034] FIG. **24**B is a diagram showing a second part of a flow chart showing a method implemented in accordance with one embodiment of the invention.
[0035] FIG. **24**C is a diagram showing a third part of a flow chart showing a method implemented in accordance with one embodiment of the invention.
[0036] FIG. **24**D is a diagram showing a fourth part of a flow chart showing a method implemented in accordance with one embodiment of the invention.
[0037] FIG. **24** is a diagram showing how the diagrams of FIGS. **24**A, **24**B, **24**C and **24**D can be combined to form a complete flow chart showing steps of an exemplary embodiment.
[0038] FIG. **25** illustrates an exemplary pallet with a payload, i.e., set of boxes, to be moved.
[0039] FIG. **26** illustrates the exemplary pallet with payload of FIG. **25** with a 3D shape, corresponding to outer edges of the pallet with the payload, said 3D shape having been determined and being displayed around the perimeter of the pallet and payload.
[0040] FIG. **27** illustrates the exemplary pallet with payload of FIGS. **25** and **26** with some of the boxes on the pallet having shifted during the move and now extending beyond and above the right side of the pallet.
[0041] FIG. **28** illustrates the exemplary the pallet with payload of FIG. **27** that shifted position during transport with a 3D shape, corresponding to outer edges of the pallet with the payload, said 3D shape having been determined and being displayed around the perimeter of the pallet and payload.
[0042] FIG. **29** illustrates a pallet rack with multiple locations in which a pallet with goods can be stored.

[0043] FIG. **30** illustrates the pallet rack of FIG. **29** but with the space available in the top two storage locations based on 3D measurements, e.g., stereoscopic camera based measurements or other measurements with the determined available spaced being represented by 3D objects in the form of cuboids shown using dashed lines.
[0044] FIG. **31** is an example of an image of the pallet rack of FIG. **29** with a pallet and goods being checked for possible insertion into the top left storage slot with the previously determined cuboid, corresponding to the pallet with goods and the cuboid corresponding to the available space, being superimposed on an image of the pallet with goods and pallet rack to allow an operator to visually understand and check whether the pallet with goods will fit in the available storage space.
[0045] FIG. **32** is a diagram showing a visual indication, e.g., color represented in the Figure as a solid black line, around the cuboid corresponding to the pallet with goods with the color indication having been added to the right side to the pallet with goods shown in FIG. **31** to visually indicate a predicted area of collision which is expected to occur if the pallet with goods is inserted into the upper left storage location of the pallet rack.
[0046] FIG. **33** shows the pallet with goods, 3D cuboid shape serving as a pallet with goods volume indicator and another cuboid serving as an available space indicator while are all visible as part of a check to make sure the pallet and goods can be successfully placed into the top right storage location of the pallet rack.
[0047] FIG. **34** shows the pallet with goods after successful insertion into the top right storage location of the pallet rack.
DETAILED DESCRIPTION
[0048] FIG. **1** is a drawing of a system **100** implemented in accordance with one exemplary embodiment. The system **100** includes a centralized operator control center **102**, one or more operator residences **104**, and a plurality of warehouses **106** through **108** which are coupled together for communications purposes via communications links **112**, **114**, **116**, **118**, **120** and a network **110**. While two warehouses **106**, **108** are shown, the centralized operator control center **102** may provide services to a large number of warehouses some of which may be located miles away or even in different states. The communications network **110** may be, and sometimes is, the Internet.
[0049] While the invention is being explained in the context of an example where the devices to be controlled are robotic forklifts located at warehouses, it should be appreciated that the methods and apparatus described herein relating to the remote control of devices which are moveable or have moveable elements such as forks or arms, can be applied to a wide range of devices that can be controlled remotely including different types of vehicles, assembly robots, etc. located in a wide range of different environments. Accordingly, it should be appreciated that the forklift related embodiment is exemplary, and that the invention is not limited to being used solely with forklifts and can be used with a wide variety of controllable devices that can be remotely controlled.
[0050] The centralized operator control center **102** is shown located remote to the warehouses **106**, **108** but may be located in, or adjacent to, one of the warehouses **106**, **108** while being remote to other ones of the warehouses **106**, **108**. The centralized operator control center **102** includes a plurality of operator workstations **122**, **124** which are coupled via corresponding links **112**, **114** to the network **110** and thus can communicate with devices **130**, **132**, **130′**, **132′** located at the various warehouses **106**, **108** via the network **110**.
[0051] Operator residence **104** includes an operator home workstation **126** which is the same or similar to the workstations **122**, **124** located at the centralized operator control center **102** allowing an operator to work from his/her home and remotely operate robotic or other controllable devices **130**, **132**, **130′**, **132″**. Operator home workstation **126** is coupled to network **110**, via communications link **116**, through which it can communicate with devices to be controlled. While a single exemplary operator residence **104** is shown in FIG. **1**, multiple operator residences may be present in the system **100**. Human operators are at the workstations **122**, **124**, **126**. As will be discussed below the operator workstations **122**, **124**, **126** include network interfaces through which

the workstations can receive signals, video and other information from robotic devices **130**, **132**, **130′**, **132′** and send control and other signals to the robotic devices **130**, **132**, **130′**, **132′**

[0052] Each warehouse **106**, **108** includes a network interface **128**, **128′** which supports a wired and/or fiberoptic connection **118**, **120** to network **110** and also supports wireless communications using antennas. The network interfaces **128**, **128′** can be, and sometimes are, implemented as WiFi access points or other local wireless communications devices which have a wired and/or wireless interface and which can support communications between the robotic devices **130**, **132**, **130′**, **132′** and network **110**.

[0053] Robotic devices **130**, **132**, **130′** and **132′** can be controlled via the remote operator workstations **122**, **124**, **126** and communicate to the operator workstations **122**, **124**, **126** via a wireless connection to the network interface **128**, **128′** in the warehouse **106**, **108** in which the device **130**, **132**, **130′ 132′** is located and network **110**. The latency of the communications connection between an operator workstation **122**, **124**, **126** and a robotic device **130**, **132**, **130′** and **132′** will depend on the latency of the individual links which connect the devices and can vary over time, e.g., based on operator workstation location, robotic device location and/or network issues which can affect latency through the network **110**. The latency can be measured in a variety of ways, e.g., with a workstation sending a signal and then measuring a time between when the signal is sent and a response to the signal is received. Similarly, a robotic device can measure the latency between the robotic device and an operator workstation controlling it by sending a signal to the workstation and then measuring the time between when the signal was sent and a response was received. The latency in one direction may be, and sometimes is, assumed to be half the round trip latency. Other techniques for measuring latency can also be used.

[0054] The amount of bandwidth that can be supported from a robotic device and an operator workstation can depend on wireless conditions at the warehouse at which the device to be controlled is located, network loading and a variety of other issues. Testing of how much bandwidth can be supported can be determined by sending a test load or other known amount of data over the communications path to be tested and the device receiving the data reporting back how much data was received in a test period of time. For example, robotic device 1 **130** can communicate test data to operator work station **122** in different known amounts and the operator workstation **122** can report back based on the received test data the amount of data, and thus bandwidth, the data path between the robotic device **130** and operator workstation **122** reliably supports. The latency and bandwidth of a communications path can be tested/measured repeatedly at different points in time and adjustments in control parameters and/or what information, e.g., video data, is communicated modified in response to changes in the latency and/or bandwidth between the robotic device being controlled and the workstation used to control the device.

[0055] As will be discussed below, the operator workstations **122**, **124**, **126** can operate as device simulators with control and other inputs corresponding to the device to be controlled and simulated outputs, e.g., sounds and/or a displayed field of view. For example, the operator workstation **122** may, and sometimes does, include one or more pedals used to control forklift motion and levers/controls used to control lifting or lowering of forklift forks and/or forklift mast tilt controls.

[0056] FIG. **2** illustrates an exemplary operator workstation **200** that can be used as any of the operator workstations **122**, **124**, **126** shown in FIG. **1**. The operator workstation **200** includes a first wired or optical network interface **206** in addition to a second wireless interface **204**. The first network interface **206** includes a receiver **218** and a transmitter **220** coupled to fiberoptic or wired link **222** which couples to the operator workstation **200** to, e.g., communications network **110**. The second wireless interface **204** includes a transceiver **224** that includes a wireless receiver **226** and a wireless transmitter **228** which are coupled to one or more antennas **230**, **232** by antenna control network **229**. The antenna control network **229** can alter antenna connectivity depending on whether receive, transmit and/or both receive and transmit operations are to be performed and whether beam forming is supported. In cases where beam forming is not supported a single antenna

230 may be used. Thus, an array **230**, **232** of antennas is optional and not used in all embodiments.
[0057] The first and second network interfaces **206**, **204** are coupled to a processor **202**, assembly of hardware components **214**, memory **212** and I/O interface **208** via bus **216** allowing the components coupled to the bus **216** to exchange information, video data, control signals, etc. The processor **202** controls the operator workstation **200** to implement the methods described herein and to perform the steps described in the present application which are performed by the operator workstation under the direction of the control routine **270** stored in memory. In addition to the control routine **270**, the memory **212** includes an assembly of components, e.g., an assembly of software components **272**. The software components **272** include processor executable instructions which when executed by the processor **202** cause the processor **202** to implement one or more functions performed as part of the methods described herein. The memory **212** also includes data/information **274** such as received video data **277**, e.g., images, of an environment in which a controlled device is working, control signals/instructions **279** obtained from operator input, in addition to communications path latency and/or bandwidth information **281**, which indicates the latency and/or bandwidth associated with a device being controlled from the operator workstation **200**. The data/information **274** also includes information **283** indicating a task to be performed by the device being controlled by the operator of the operator workstation **200**. The task may be indicated by the operator of operator workstation **200** or in many cases may be signaled to the operator workstation **200** by a robotic device when the device initiates an automatic request for operator control or operator assistance when a control device determines that an operation scheduled to be performed is risky and should not be implemented automatically without operator assistance. This is particularly the case with some semi-autonomous devices where a robotic device will perform some low risk operations automatically but automatically connect to an operator workstation when a risky operation such as picking up a pallet of items or working near an edge with a fall off is required. The task information **283** can also be automatically inferred by the processor **202** detecting a known sequence of operations corresponding to a particular task which is determined and then stored in memory **283**. For example, a known operator initiated sequence of operations corresponding to approaching and picking up a loaded pallet may result in the operator workstation **200** or device being controlled determining that a loaded pallet pick up task is to be performed.
[0058] Various input and/or output devices are coupled to the I/O interface **208**. In the FIG. **2** example these input and/or output devices includes a microphone **240**, speaker **242**, camera **244**, switches **250**, mouse **252**, keypad **254**, joystick **256**, control status panel **258** and one or more displays **246**, **248**. While some of these devices such as the displays **246**, **248** can be used for device control related operations such as displaying video received from or corresponding to the device being controlled, in some embodiments device operator simulator components are also included to provide the operator an experience similar to what might be encountered if the operator were directly operating the device, e.g., forklift, at the site where the device is located. Device, e.g., forklift, simulator **260** is coupled to the I/O interface to provide device inputs which are the same or similar to those an operator of the device would use to control the device, e.g., forklift, if present at the device. The simulator **260** includes various on/off control switches **262** used for enabling/disabling different device functions, a motion, e.g., speed, control device **264** such as a fuel pedal, one or more steering control devices **266**, e.g., a steering wheel, and/or controls **268** for device attachments such as a forklift mast tilt control and/or fork raise/lower control. Input signals generated by the device simulator **260** are automatically converted into electrical signals and/or commands which are supplied to the I/O interface and thus the processor **202** via bus **216**. The simulator **260** allows an operator to control the remote device while images of the device environment are displayed on one or more of the displays **246**, **248**.
[0059] As will be discussed below, the maximum speeds permitted at different times and/or the rate at which components of the device being controlled are allowed to move can be, and sometimes

are, made a function of communications path latency. For example, the turning of the steering control device may result in a slower response and/or require more rotations in the case of higher latency than when there is low latency. In this way the operator controls automatically reflect to some extent the latency in the communications path. Similarly, the maximum speed in a given direction or the rate at which forks are raised/lowered may be, and sometimes are, scaled or limited based on latency. The slowing of speeds as communications latency increases gives changes in position more time to be captured on camera or detected by another sensor and communicated back to the operator for display. Thus the operator is less likely to make positional control errors due to a miss-understanding of device position even in cases where latency changes. Device speed and/or other control operations may be, and sometimes are, controlled based on the task to be performed and/or environmental conditions in addition to latency. While communications latency is used in some embodiments as a control factor, in some embodiments control parameters are adjusted based on environmental conditions and/or the task to be performed even if latency is not considered.

[0060] In some embodiments, the image or image portions of the environment displayed may be, and sometimes are, selected/controlled based on the amount of bandwidth available between the operator workstation **200** and/or the task to be performed.

[0061] FIG. **3** illustrates an exemplary device **300**, e.g., a robotic forklift, which can be remotely controlled in accordance with some embodiments and which can be used in the system of FIG. **1** as any one of the robotic devices **130**, **132**, **130′**, **132′** shown in FIG. **1**. The forklift **300** includes a main housing **301** in which a motor and other components are housed or connected to, wheels **327**, **329** a mast **395** which in various embodiments can be controlled to tilt and a set of forks **323**, **325** and which sometimes serves as a camera array mounting location, e.g., a location on which camera array **392** can be and sometimes is mounted high up. An extendable fork assembly support **397** attached the fork assembly including backrest **321** and forks **323**, **325** to the mast allowing the fork assembly to be moved extended and retracted in combination with tilting of the mast **395** as needed. The backrest **321** in some embodiments includes a vertical center member on which a pallet position sensor **305** is mounted and multiple horizontal members on which the forks **623**, **625** can hang. The backrest **321** may further include a railing or guard behind the horizontal members which extends up above the horizontal member and reduces the risk of a load falling in the direction of the forklift mast **610** as shown in FIG. **6** for example.

[0062] In the FIG. **3** example an obstacle detected indicator light bar **307** which includes a string of independently controllable lights is included along a top portion of the robotic device, e.g., along the top of the main body **301** or a canopy which is used in some embodiments. In some embodiments a light or lights on the light bar **307** closest to a detected obstruction are illuminated to indicate that the obstruction was detected. The lights of the lightbar **307** extend around the robotic device to provide a range of indicators extending 360 degrees around the robotic device. FIG. **3** is simplified diagram and additional examples of obstruction detected indicator light bars are visible in FIGS. **6-10** which show alternative embodiments of the exemplary robotic device shown in FIGS. **3** and **4** and which can include the components of the robotic devices shown in FIGS. **3** and **4** but perhaps in a slightly different arrangement and/or with different placements for various components.

[0063] The forks **323**, **325** can be raised and lowered. The robotic device **300** includes a variety of cameras and other sensors as will be discussed further with regard to FIG. **4**. In the FIG. **3** diagram, a first camera system **392**, which is a forward looking camera system (FLCS), which includes a first set of forward looking cameras is shown, and a second camera system **394**, which is a rear looking camera system (RLCS), which includes a second set of rear looking cameras is also shown. The FLCS **392** provides images of the forklift area to facilitate an operator controlling a pallet pickup/lowering operation and/or other pallet manipulation as well as controlling forward travel of the forklift **300**. In some cases, the FLCS **392** and RLCS **394** are or include sets of cameras which provide images that are used for stereoscopic depth determination and/or obstruction detection with different images of the same scene area being provided by different cameras in the area to facilitate stereoscopic depth and 3D shape determinations. Side facing pairs of cameras and/or other sensors may be, and sometimes are, also included, allowing obstructions to be detected in all directions. The stereoscopic cameras and/or other sensors also support determining the distance to detected obstructions and/or determining the type of detected obstruction, e.g., human or another type of object. The cameras of RLCS **394** are used to provide a view of the area behind the forklift device for backup or other operations. While shown as mounted on the body of the robotic forklift in some cases the cameras of the FLCS **392** or another set of forward looking cameras are positioned on the backrest **321**, which is part of the fork assembly which also includes forks **323**, **325**.

[0064] In some cases, a pallet position sensor **305** is also mounted on the backrest **321** of the fork assembly allowing a determination to be made as to when a pallet is nested, e.g., seated against, the backrest **321**. The position sensor **305** is able to measure the distance from the sensor **305** to a pallet on the forks **323**, **325** and this distance information is used to determine, in some cases, if the pallet is positioned against the backrest **321** to which the position sensor **305** is mounted. To avoid damage to the position sensor **305**, the position sensor **305** is sometimes mounted slightly behind the frontmost surface of the backrest **321**, which should prevent actual physical contact with the sensor **305** which may be, and sometimes is, an ultrasonic distance sensor or another type of distance sensor which does not require physical contact to measure distance. In other embodiments a contact sensor may be used to determine that the pallet is properly positioned, e.g., at the proper distance from where the main body of the sensor **305** is mounted however such distance position sensors are more likely to be subject to damage from physical contact with a pallet.

[0065] In the case where the bandwidth between the forklift **300** and operator workstation **122** used to control the forklift **300** supports the communication of multiple images, multiple views, e.g., RLC and FLC video, will be supplied to the operator workstation **122** to provide a large amount of situational awareness. In cases where the available bandwidth will not support multiple views, one of the camera outputs may be prioritized, while feeds from the other camera outputs are discontinued or transmitted less frequently than from the highest priority camera. In some embodiments foveation is supported with portions of a camera output, images, being processed, e.g., subject to cropping and/or low resolution representation, based on what portion of a scene the operator working at the operator workstation **122** selects to view at a given time. The viewing, e.g., field of view information, is detected using a sensor, e.g., eye tracking sensor, at the workstation **122** and communicated to the device being controlled to facilitate foveation and/or selection of which camera feed or feeds to supply to the operator workstation **122** when bandwidth is limited.

[0066] Having generally described the device, e.g., robotic forklift, **300** to be controlled with regard to FIG. **3**, it will now be discussed in greater detail with reference to FIG. **4**.

[0067] FIG. **4** is a block diagram of the exemplary device **300** of FIG. **3** showing various components of the exemplary device **300** in block diagram form and in greater detail than is shown in FIG. **3**. The components of the device **300** shown in FIG. **4** are included in the robotic devices shown in the other figures and examples of this application as well in some embodiments and thus FIG. **3** is to be considered exemplary of the components which are included in various robotic device embodiments.

[0068] Controllable device **300**, e.g., a robotic forklift or another device, includes an input device **302**, an output device **304**, an obstacle detected indicator light bar **307**, a pallet position sensor **305**, a processor **306**, an assembly of hardware components **308**, e.g., an assembly of circuits, memory **310**, a communications interface (COM interface) **311**, a navigation/guidance system **340**, a vehicle control system **344**, a plurality of sensors (sensor 1 **370**, . . . , sensor n **372**), a plurality of sonar units (sonar unit 1 **374**, sonar unit 2 **376**, . . . , sonar unit K **378**), a plurality of radar units (radar unit 1 **380**, radar unit 2 **382**, . . . , radar unit L **384**), a plurality of LIDAR units (LIDAR unit 1 **386**, LIDAR unit 2 **388**, . . . , LIDAR unit M **390**), a plurality of cameras systems (camera system 1 **392**, which is a FLCS including a set of forward looking cameras, camera system 2 **394**, which is a

RLCS including a set of rear looking cameras, . . . , camera system N **396** including another set of cameras), and an artificial intelligence (AI) unit **313** coupled together via bus **309** over which the various elements may interchange data and information. While a combination of sensors, sonar, radar and LIDAR units are shown in the example, the device **300** may include some or none of these sensors/measuring devices depending on the embodiment. The communications interface **311** includes a wireless interface **312** and a network interface **314**. The wireless interface **312** includes a receiver circuit **316** and a transmitter circuit **318**. The network interface **314** includes a receiver circuit **317** and a transmitter circuit **318**. The robotic device **300** can communicate with the operator workstation using either the wireless interface **312** and/or network interface **314**. In the case of wireless communications, the communications may occur directly with the operator workstation wirelessly but in many cases wireless signals are communicated between a warehouse network interface **128** such as a WiFi access point and then between the WiFi access point and operator workstation via one or more wired or optical connections.

[0069] Input device **302**, e.g., a keypad, is used for receiving manual input, e.g., from a service technician, e.g., to alter device settings and/or access information stored on the device, e.g., robotic forklift. The output device(s) **304** e.g., a display and/or status lights or indicators and/or an audio output device, e.g., alarm, siren, speaker, etc., is used for outputting information, warnings, and/or status indications to an operator of the device **300**, a device monitoring individual, or a device service person. The processor **306**, e.g., a CPU, executes routines, e.g., routines loaded into the processor **306** from memory **310** to control the operation of the robotic device **300**. A video compression and/or selection module **333**, included in the assembly of software components **332**, controls video selection and/or compression and thus in some cases controls what video or images will be returned to the operator workstation controlling the device **300**.

[0070] Memory **310** includes assembly of software components **332**, e.g., routines, and data/information **334**. In some embodiments, a component, e.g., a routine, in the assembly of software components **332**, when executed by processor **306**, **399**, or **361**, implements a step of an exemplary method, e.g., the method of flowchart **500** of FIG. **5**. The assembly of software components **332** includes the video compression module and/or selection module **333**. The assembly of components **332** also includes an obstacle detection and indicator routine or module **315** which detects obstacles, e.g., based on sensor input, determines distance to individual detected obstacles, the type of detected obstacle and/or which light or set of lights on the light bar **307** should be illuminated to indicate that a particular obstacle was detected. FIG. **5** illustrates steps which may be, and sometimes are, implemented by the robotic device **300** under control of processor **399** when obstacle detection and indicator routine **315** is implemented, e.g., during autonomous mode or semi-autonomous mode operation or during other modes of operation, e.g., manned mode of operation where automatic obstacle detection is implemented. Pick control routine and/or component **331** controls pick operations, e.g., pallet pickup operations, based on the distance from the forklift body position to a pallet to be picked up and/or based on the output of pallet position sensor **305**. FIG. **10** illustrates steps which can be and sometimes are implemented under control of pick control routine **331** when it is executed by a processor such as processor **399**. In some embodiments pick control routine **331** will trigger a double bite pickup operation when a pallet is too far for the forklift to reach and achieve a nesting position from where the forklift can park to perform the pickup. In other cases, an attempt to implement a pallet pickup and achieve nesting in a single pickup operation may fail for a variety of reasons including slight inaccuracy of robotic device placement at the start of the pick operation. In response to the pallet position sensor **305** detected that a pallet has not been positioned in the desired nesting position against the backrest the pick control routine **331** will trigger a place and repick operation which normally results in proper nesting of the pallet before the robotic forklift begins moving to a new location.

[0071] Data/information **334** includes information **337** including received video feed(s) and/or information, e.g., sensor information, captured video and/or information, e.g., sensor information, from one or more of the cameras/sensors. The memory **310** includes operating parameter information such as maximum device speed and/or acceleration and/or max speed and/or acceleration of components, e.g., forks, which are part of the robotic device **300**. Memory **310** includes a set of predetermined parameters **341**, e.g., max speed and/or max acceleration for different tasks which may be, and sometimes are, performed by the robotic device. For example, the predetermined parameters **341** may specify a first maximum speed for the robotic device **300** moving between locations with items on its forks and a second lower maximum speed for the robotic device **300** moving with one or more items on its forks.

[0072] In addition, the parameters may include a maximum acceleration for one, more or all tasks for which a maximum speed is indicated. Thus, in some embodiments the predetermined parameters include pairs of maximum permitted speed and maximum permitted acceleration for multiple tasks which can be performed. Tasks for which maximum speed and acceleration are specified can include tasks relating to movement of a device component such as forks of the robotic device **300**. In one embodiment the predetermined parameters include a first predetermined maximum fork raising and/or lowering speed, and corresponding maximum fork acceleration, for the task of raising or lowering the forks when the forks are empty and a second lower predetermined fork raising and/or lowering speed, and corresponding second maximum fork acceleration, for the task of raising or lowering the forks when the forks are loaded, e.g., with a pallet or other item(s). The predetermined parameters may be, and sometimes are, maximum speeds/accelerations which can be safely supported for various tasks likely to be performed by the robotic device **300**, e.g., under operator control, in the absence of communications latency problems or environmental hazards such as the presence of people or drop offs such as stairs, pot holes or the edge of a loading dock off of which the robotic device **300** might fall.

[0073] Determined parameters **343** are operating parameters, e.g., determined by processor **306**, that the robotic device should use to perform a determined task based on communications latency and/or environmental conditions present at the time the task is to be performed under control of a remote operator workstation **122**. The determined parameters **343** include maximum device speed and acceleration and/or maximum speed of a device component, e.g., forks and/or corresponding maximum acceleration. The determined parameters **343** will in many cases be lower than the predetermined parameters for the task to be performed because they may and often will have been modified, e.g., lowered, based on communications latency between the robotic device **300** and operator workstation controlling the device and/or because of the presence of one or more environmental hazards such as the presence of a drop off or a human in the robotic device's operating area, e.g., the environment in which the robotic device is to perform the task for which the control parameters **343** are determined.

[0074] The memory **310** also includes information **338** on a determined communications latency and/or bandwidth between the controllable device **300** and the operator workstation **122** being used to control the device **300** and task information **339** indicating a task to be performed by the device **300**. The task to be performed, may be determined based on stored instructions or e.g., by the artificial intelligence unit **313** based on a sequence of operations initiated by a remote operator.

[0075] Wireless interface **312** includes a wireless receiver **316** coupled to receive antenna **320** via which the device to be controlled, e.g., forklift **300**, receives control signals, commands from the operator workstation. The wireless interface **312** also includes a wireless transmitter **318** coupled to transmit antenna **322** via which the device **300** transmits wireless signals, e.g., sensed information and/or video, along with task information in some cases. In some embodiments, there is a wireless link between the wireless interface **212** and a network interface **128** or **128′**. The network interface, e.g., network interface **128** is responsible for acting as an intermediary and for communicating signals sent using network **110** to/from the controllable device **300**. Network interface **314** includes a receiver **317** and a transmitter **319**, via which the device **300** can communicate with test equipment and/or which is used to configure the device **300**, e.g., when the device is parked, e.g., at a depot, garage, and/or service, repair and/or maintenance facility into which a wired or fiber connection can be made with the network interface **314**.

[0076] Controllable device **300** further includes an embedded GPS receiver **338** coupled to GPS antenna **330**, which receives GPS signals and determines time information, a position fix, e.g., latitude/longitude/altitude, and/or velocity information for the device **300**. The output of GPS receiver **338** is fed as input to navigation/guidance system **340**. Navigation/guidance system **340** includes an inertial measurement unit (IMU) **342**, e.g., an IMU on a chip, including gyroscopes and accelerometers. Navigation/guidance system **340**, provides filtered location, attitude, acceleration and/or velocity information, and is used to route the device **300**, e.g., a forklift, along an intended path. While GPS is used when GPS signals are available, in indoor situations the device **300** may operate without GPS signals/information.

[0077] Device control system **344** includes a fork position control system **352**, a power control system **354**, a steering control system **356**, a braking control system **358**, and processor(s) **361**. Power control systems **354** is coupled to motor/engine/transmission/fuel system(s) **346**, included in device **300**, e.g., a forklift **300**, and is used to control motion of the device **300**, e.g., forklift **300**, e.g., forward, reverse, speed, acceleration, deceleration, etc., motion of the forks **323**, **325** and/or fork mast **395** which in some embodiments can be tilted. Steering control system **356** is coupled to directional control components **348**, e.g., steering motors, steering linkages, rack and pinion, gear box, etc., and is used to control the direction of device **300**, e.g., forklift **300**. Braking control system **358** is coupled to braking components **350**, e.g., brake actuators, brake cables, wheel speed sensors, wheel motors, ABS system components, etc., included in device **300**, e.g., forklift **300**, and is used to control braking of the device, e.g., forklift **300**.

[0078] Sensors (**370**, . . . , **372**) include, e.g. speed sensors, motion sensors, proximity sensors, etc., and collect information used to control device **300**, e.g., forklift **300**. Sonar units (**374**, **376**, . . . , **378**) are used to perform distance measurements to other objects in the vicinity of the device **300**, e.g., forklift **300**. Radar units (**380**, **382**, . . . , **390**) are used to detect and measure the speed of other objects in the vicinity of the device **300**, e.g., forklift **300**. Light detection and ranging (LIDAR) units (**386**, **388**, . . . **390**) use lasers or other light to detect and range objects in the vicinity of the device **300**, e.g., forklift **300**. Cameras included in the camera systems (**392**, **394**, . . . **396**), are mounted and oriented on the device **300** to capture different fields of view, capture images, e.g., video feeds, which are streamed, e.g., selectively streamed, to the operator control station **122**, and/or are used to provide assistance in automatically moving or repositioning the device **300**, e.g., forklift **300**, e.g., when performing some tasks in a full automated manner.

[0079] Video feed processing **393** receives video feeds from cameras on the device **300**, e.g., forklift **300**, and provides selection, compression, cropping and/or foveation on captured images, e.g., images of the environment and/or scene area in which the device, e.g., forklift is being used, prior to the images being communicated to the remote operator. Processing performed by video feed processing unit **393** can be, and sometimes is, based on information about the scene area an operator is looking at which is detected at the operator workstation **122** and communicated to the controllable device, e.g., forklift **300**, in some embodiments.

[0080] FIG. **5** is a flow chart **500** illustrating the steps used in one exemplary embodiment to provide a visual indication of one or more detected obstructions which may be, and sometime are, objects or human beings in the area of a robotic device, e.g., an autonomous, semi-autonomous, or remotely controlled material handling vehicle such as the robotic device **300**. The method starts in step **502** with the robotic device and the processor in the robotic device being powered on. Operation proceeds from start step **502** to step **504**, in which sensor data is collected. This can, and sometimes does, involve receiving images from cameras, output of one or more Lidar sensors, the output of one or more ultrasonic sensors, or other types of sensors which are used to sense objects, e.g., obstructions, in the area of the robotic device. The sensor data collected in step **504** is processed in step **506** to identify obstacles in the vicinity of the robotic device. Individual obstacles may be, and sometimes are, detected in step **506**. Obstacles detected in step **506** may be, and sometimes are, objects such as boxes, racks, curbs and/or human workers in the area of the robotic device.

[0081] In step **508** a check is made to determine if any obstacles were detected in step **506**. If no obstacles were detected, operation proceeds to step **509**. In step **509** obstacle indicator lights on the robotic device are set to indicate that no obstacle has been detected, e.g., the lights on the obstacle indicator light bar are turned off or set to a color that indicates no obstacles have been detected. Operation process from step **509** back to sensor step **504**, which is repeated on an ongoing basis so that changes in the environment and/or changes in vehicle position can be taken into consideration and obstacle light bar indicator output updated accordingly.

[0082] Referring once again to step **508** if in step **508** one or more obstacles were determined to have been detected, operation proceeds to step **510**. Steps **510** to step **520** will be performed for each detected obstacle.

[0083] In step **510** the distance, e.g., from the closest side of the robotic device to the detected obstacle, is determined. Then in step **512** the number of lights to be illuminated is determined based on the distance. For example, in some embodiments the closer the obstacle is to the robotic device, the greater the number of lights facing in the direction of the object that will be illuminated. Thus, the number of lights in some cases is inversely related to, e.g., inversely proportion to, the distance to the detected obstruction, with the shorter the distance the greater the number of lights to be illuminated. In some embodiments the determined number of lights is a group of physically adjacent lights which are determined to correspond to the detected object for purposes of signaling detection of the obstruction.

[0084] Operation proceeds from step **512**, in which the number of lights to be illuminated is determined based on distance, to step **514**, in which the orientation, e.g., direction, of the detected obstruction (obstacle) relative to the robotic device, e.g. relative to the front of the robotic device, is determined. This information is used to determine which lights correspond to the direction of the detected object (obstacle) and thus which lights should be illuminated to signal the direction of the obstruction (obstacle).

[0085] Operation proceeds from step **514** directly to step **518** in cases where optional step **516** is not performed or via step **516** in case where this obstacle type determination step is performed. In step **516** a determination is made as to the type of obstacle that was detected, e.g., an inanimate object or a human being. Various known object recognition routines can be used to implement this step which is used in determining what color light to activate in cases where the type of object that was detected is to be visually signaled in addition to the fact that the obstruction was detected.

[0086] In step **518** the color, hue and/or intensity of lights in a group of lights is determined to correspond to the detected obstacle for signaling purposes, e.g., based on the number and direction information determined in steps **512** and **514**, based on the obstacle type determined in optional step **516**, if performed, and/or based on default setting information, is determined. The lighting determination made in step **518** allows the group of lights to effectively point in the direction of the detected obstruction, e.g., by having the center light of the group of lights to serve as a pointer and be the brightest or be of a particular color and also, in some embodiments, to signal the type of object which was detected.

[0087] Operation proceeds from step **518** to step **520** in which the group of lights determined to correspond to the detected obstruction (obstacle) are illuminated based on the color, hue and/or intensity determination made in step **518**, e.g., with different lights in the group of lights having different colors, hues and/or intensity as appropriate to communicate the determined information (e.g., including direction of obstacle and/or type of obstacle) with the number of lights illuminated being used to indicate the distance to the detected obstruction (obstacle).

[0088] With a group of lights corresponding to a detected obstruction (obstacle) having been activated in step **520**, e.g., to provide illumination signaling detection of the obstruction, operation proceeds to step **522**. In step **522** a check is made to determine if additional obstacles were detected, e.g., requiring the illumination of more lights. In step **522** if it is determined that additional obstacles were detected, operation proceeds back to step **510** so that steps **510** through **520** can be performed for each obstacle detected in step **506**.

[0089] If in step **522** it is determined that no additional obstacles were detected, e.g., that steps **510** through **520** were performed for each detected obstacle, operation proceeds to step **524**. In step **524** previously activated obstacle indicator lights which no longer correspond to a detected obstacle are deactivated. In this way when a previously detected obstacle is no longer detected, e.g., because the robotic device has moved away or the obstacle, e.g., human being, which was previously detected has moved away, the lights used to signal detection of the obstacle which is no longer present will be deactivated if they are not part of a group used to signal the detection of some other obstacle.

[0090] Operation proceeds from step **524** to step **504**, indicating that obstacle detection and visual indicating of detected obstacles will be performed on an ongoing basis. As a result, the output of the obstruction detection indicator lights will change over time as the environment in which the robotic device changes and/or the robotic device moves in the environment resulting in changes in which obstructions are detected and/or their position relative to the robotic device.

[0091] As should be appreciated from the description of FIG. **5**, a light array **307**, e.g., string or bar of lights, used as obstacle detection indicators, are arranged on one or more sides of a robotic device, e.g., in a row at one or more locations on the body of the robotic device. The robotic device may be, and sometimes is, an autonomous, semi-autonomous, or remotely controlled material handling vehicle, e.g., a robotic forklift or semi-autonomous forklift which can be remotely controlled under some circumstances. Lighting signals are used to indicate the detection of specific obstacles seen by the robotic device implementing the invention.

[0092] FIGS. **6-8** show exemplary robotic forklifts **600**, **800** that are used in some embodiments with the forklifts being shown from different perspectives in different figures. In FIG. **6**, the robotic forklift **600** including a main body **602**, mast **610**, canopy **620**, fork backrest **321** and forks **623**, **625**. Forward looking camera system **392** and pallet position sensor **305** are clearly visible in FIG. **6** and are mounted to the center support of backrest **321**. The forklift **602** may include the same components as the robotic device of FIG. **4** but with the combination of lights **607′**, **607″**, **607‴** (see FIG. **7**) serving as the light bar **307** of the FIG. **4** example. Note that in the FIG. **6** example the obstacle detected indicator light bar **607**, which includes light strings **607′**. **607″**, **607‴** are mounted on the canopy **620** so that they are clearly visible to people in the area of the robotic device **600**. In FIG. **7** the rear of the robotic device **600** can be seen with the lights **607‴** on the canopy **620** being clearly visible. Thus, the lights **607** are present on all sides of the robotic device and can be used to signal detection of an obstruction in any direction by illuminating a light on the light bar **607** corresponding to the direction of the detected object relative to the robotic device **600**. Door **621** allows a human to access the main body **602** of the robotic device for control and/or component access purposes.

[0093] FIG. **8** illustrates another exemplary robotic device **800** with a light bar **807** including light strings **807′**, **807″** and **807‴** (see FIG. **9**) and **807⁗** (see FIG. **9**) being used as the obstacle detection light bar identified in FIG. **4** as reference number **307**. Note that in the FIGS. **8** and **9** example the lights of the obstruction detected indicator light bar are positioned on the main body **802** and door **821** leaving the canopy **820** light free or available for other lights, e.g., a siren type light and/or lights used to signal autonomous vs remote controlled operation. Note that in FIG. **9** the lights extend across the top of the rear door **821** since the body **802** is interrupted by the door **821** at this location.

[0094] The lighting implementation used in various embodiments such as the ones shown in FIGS. **6-9** provides a feedback mechanism for those observing the vehicle **600** or **800**, e.g., human workers in the area of the vehicle or human observers remotely monitoring the vehicle by the use of one or more cameras, to be made aware of the obstacle detection status of the robotic device.

[0095] Light is emitted from the robotic device **600** or **800**, e.g., via lights in the light bar **607** or **807** facing the direction of a detected obstacle to signal that the obstacle has been detected and the determined distance to the detected obstacle. In this way the obstacle detector indicator light bar **607** or **807** provides an observer a visible indicator that the system is aware of the presence of the detected obstacle when other methods would not be practical or appropriate to notify the observer that the obstacle has been detected. In at least some cases the obstacle is a human worker in the area of the robotic device **600** or **800**.
[0096] In some embodiments the type of obstacle which is detected is determined, e.g., using known object recognition algorithms, and the light is controlled as a function of the type of detected object, e.g., a different color can be used to signal detection of a human than an inanimate object.
[0097] The indication of a detected object through the use of one or more lights facing in the direction of the detected obstacle, e.g., object, provides a visual signal that can be easily perceived by a human in the area of the robotic device in a similar way that visual signs from a human operator can signal that a human operator is aware of the person working in the area. Thus, the invention mimics the use of visual indication of obstacle detection by a human vehicle operator, through the use of directional lighting highlighting which obstacles are in view, where they are being observed, and how far away they are from the vehicle without the need for a human operator to be involved in the signaling of such information.
[0098] While visual indicators of obstacle detection can be useful to signal detection of an obstacle, the lack of a visible obstacle indication, when an observer can see an obstacle in the area of the robotic device, allows the observer to determine that the robotic device's obstacle detection system has failed or may not be working properly. Thus, the methods are not only useful in signaling obstacle detection but can also be useful in identifying or detecting problems with the obstacle detection system.
[0099] In some embodiments a light bar (e.g., an LED light bar) **307**, **607** or **807** is used as an obstacle indicator device. The light bar in some embodiments is affixed to the robotic device **600** or **800**, e.g., autonomous vehicle, in a position with a clear view to observers operating around the vehicle. For example, in the case of a reach truck or other forklift device the light bar may be, and sometimes is, placed around a top cab guard and/or an enclosure providing 360 degree visibility to those around the vehicle. The light bar **307**, **607** or **807** in some embodiments is integrated into the vehicle's obstacle detection functionally and/or human detection system, e.g., routine **500** and the corresponding components, in such a way that directionality in relation to the vehicle position can be displayed in the light bar **307**, **607**, **807** when an obstacle, which in some cases could be a human, is detected.
[0100] In some embodiments as the robotic device/vehicle position and/or obstacle position change relative to each other, the lights, e.g. the LEDs, on the light bar **307**, **607** or **807** track and display the relative position between the vehicle and the obstacle in real time. The lights can, and in some embodiments do, signal both distance and bearing to a detected obstacle. Distance to an obstacle can be, and sometimes is, indicated by the number of lights illuminated as a set, e.g., with the nearer a detected obstacle is to the device the greater the number of adjacent LEDs in the direction of the obstacle which are illuminated. Direction of the obstacle can be, and sometimes is, indicated by the center of the illuminated lights, e.g. LEDs, corresponding to an obstacle being in the direction of the obstacle. The intensity, color or hue can be, and sometimes is, used to create a perceived arrow or direction indicator pointing in the direction of the detected obstacle, e.g., with the center of the arrow being made brighter than the outer lights of the group of lights being used to point in the direction of the obstacle. In some embodiments a color is used to indicate the type of obstacle that was detected, e.g., with orange being used to indicate a human and red or another color being used to indicate other objects. The particular colors used to indicate different types of objects can vary depending on the implementation.

[0101] Thus, in various embodiments the lights, e.g., LEDs, of the lights bars, represented as small circles in FIGS. **6-9**, that are illuminated to indicate detection of an object, effectively point in the direction of the obstruction communicating detection of the obstruction and acknowledging the obstructions presence. When an obstacle is no longer detected, the lighting indicator corresponding to the previously detected object will no longer show on the light bar (e.g., the LED light bar), e.g., the lights corresponding to the object will be turned off, indicating that the obstacle which was previously detected is no longer detected by, e.g., is no longer visible to the robotic device.

[0102] The number of sides of the robotic forklift on which lights are placed can be as few as one. However, to provide visibility of lights serving as obstacle detection indicators from multiple directions the light string is normally placed on multiple sides of the robotic device **300** and in some cases on all four sides of the robotic device **300**. In some embodiments the lights are placed high up, e.g., along the edge of a canopy or roof to increase visibility. The obstacle detection indicator lights may be in addition to one or more lights used for other purposes, e.g., to indicate whether the robotic device is operating in a robotic mode of operation or a remotely controlled mode of operation.

[0103] While the obstacle indicator lights are placed high up, e.g., around a canopy or roof in some embodiments, in other embodiments they may be located lower down on the body of the vehicle, e.g., mid-door height or along a middle section of the robotic device.

[0104] In some embodiments the lights extend around the upper portion of the forklift body forming what might be thought of as a halo or crown of lights. Sensors, e.g., cameras of camera arrays **394**, **392** or a processor processing the images of a camera array or another sensor, detect obstructions, e.g., human workers, near the forklift. Lights surrounding the robotic device nearest to the obstruction are identified based on the direction of the obstruction, e.g., relative to the front of the robotic device. The number of lights to be illuminated based on an individual detected obstruction, e.g., sometimes referred to as an obstacle, is in some embodiments, a function of the proximity of the obstacle to the robotic device. In some embodiments the closer the obstacle to the robotic device the greater the number of lights which will be illuminated. The number of lights to be illuminated due to the detection of an individual obstacle can be, and sometimes are, treated as a group of lights corresponding to the obstacle. The light or LED at the center of the group closest to the obstacle can be indicated using a higher intensity, a particular predetermined color or hue. By controlling the intensity, hue and/or color of lights in a group corresponding to a detected obstacle, is possible to give a human viewer the impression that the lights can, and sometimes do, point to the detected obstacle.

[0105] As discussed above with regard to various figures including FIGS. **3**, **6** and **9**, in some embodiments the robotic forklift **300**, **600** or **800** includes, a camera system **392** which includes or is a camera array, in addition to a pallet position sensor **305**. In some embodiments such as the ones shown in FIGS. **6** and **8** the pallet position sensor is mounted at a fixed location on the backrest so that the distance it measures is fixed relative to the rear of the forks allowing for the measured distance to be used to determine if the pallet is fully nested. The position sensor **305** in some embodiments outputs a predetermined signal indicating successful pallet nesting when a pallet is successfully inserted on the forks and rests against the backrest and a signal indicating that a pallet is not fully nested at other times. The position sensor **305** is positioned above the camera array **392** on the front of the vehicle on the backrest of the fork assembly in at least some embodiments such as the ones shown in FIGS. **8** and **9**. The cameras of the camera array **392** can see the forward facing area and thus provide a view the fork area. The cameras of the camera array **392** can capture images of a pallet in front of the forklift while the position/distance sensor **305** measures the distances between the sensor **305** and an object on the forks, e.g., a pallet. The camera array **392** can be and sometimes is positioned high up on the mast of the robotic forklift or an additional stereoscopic array in addition to camera array **392** is includes so that the location of the front edge of the pallet can be determined from captured images and/or images of a pallet storage area can be captured while a pallet is loaded on the forks. This allows a 3D shape of the pallet with goods on it as well as a 3D shape of a pallet rack where the pallet is to be placed to be generated from captured images even when a pallet is loaded on the forks.

[0106] The output of the sensor **305** indicates whether or not successful pallet nesting has been achieved. A processor, e.g., processor **399**, used to control forklift operation can determine if a pallet is nested, e.g., fully inserted in the forks, e.g., not fully on the forks as is the case when sensor **305** measures a gap between the rear of the forks and the pallet.

[0107] In various embodiments, if the output of the sensor **305** indicates a lifted pallet is not fully nested a double bite procedure is implemented with the pallet be moved closer to the forklift, deposited on the ground and then lifted in an attempt to properly nest the pallet prior to transport.

[0108] FIG. **10** is a flow chart showing the steps of an exemplary method **1000** for controlling a robotic device to perform a pick operation, e.g., a pallet pick up operation. The method **1000** can be, and sometimes is, used to control robotic device operation, e.g., operation of a robotic forklift **300**, **600** or **800**, based on the output of pallet position sensor **305**.

[0109] The exemplary method **1000** begins in start step **1002**, e.g., with a control device such as a processor, e.g., processor **399** or a processor of a workstation, being powered on so that it can be used to perform robotic device control operations. Operation proceeds from start step **1002** to instruction processing step **1004**, in which an instruction to move an object, e.g., a pallet with goods, is received. In response to receiving the instruction to move a pallet, operation proceeds to step **1006**, in which the location of the pallet is determined, so that a determination can be made as to where the robotic device should be positioned, e.g., parked, to perform the pallet pickup, lift, operation before the pallet is moved to a new location.

[0110] With the location of the pallet, to be picked and moved having been determined in step **1006**, operation proceeds to step **1008**, in which a safe location to position the forklift device, e.g., a location where the forklift device can be parked and held stationary for the pickup operation, is determined. With pallet location and forklift location, e.g., parking position, for the pickup having been determined, operation proceeds to step **1010** in which the forklift is moved to the location from which the pallet pickup, also referred to as a lift, operation is to be implemented.

[0111] In some embodiments operation proceeds from robotic device placement step **1010** directly to pallet pickup step **1022** while in other cases a check is made in step **1012** as to whether a double bite pick up is required due to the location of the pallet before a pickup operation is implemented. For example, in some cases the pallet may be recessed and picked up but proper nesting of the pallet on the forks can not be achieved in a single pickup operation due to how far away the pallet is from where the robotic device, e.g., forklift is parked. Movement of the robotic forklift closer to the pallet may not be possible due to a step or obstruction but the forks in such a case might still be able to be used to move the pallet closer to the forklift **300**, **600**, **800** so that a second pickup operation can be used to achieve proper nesting of the pallet on the forks prior to transport to the target location.

[0112] In embodiments in which optional step **1012** is implemented, operation proceeds from step **1010** to step **1012**. In step **1012** a check is made as to whether the pallet pickup and proper nesting against the forklift backrest **321** can be achieved in a single operation. This determination is based on whether extension of the forks **323**, **325** can achieve the desired nesting or that the forks can be inserted into the pallet to move it but can not be extended far enough to achieve full nesting in a single extension operation. If it is determined that full nesting can not be achieved with a single pickup operation, a decision is made in step **1013** to implement a double bite procedure. If in step **1012** it is determined that full nesting of the pallet can be achieved from the forklift location using a single bite pickup procedure operation proceeds to step **1022** from step **1012**.

[0113] From double bit decision step **1013** operation proceeds to step **1014**, in which the robotic device performs a pallet pickup operation. This, pickup step **1014**, is the first step of a double bit pickup operation in which the double bite pickup is being implemented because of the decision in step **1013** to implement such an operation, e.g., due to pallet location.

[0114] Operation proceeds from pickup step **1014** to step **1016**. Returning once again to step **1012**, if a decision is made that the pickup operation can be implemented in a single pickup which should be able to achieve full nesting of the pallet on the forks, operation proceeds to step **1022**, in which a pallet pickup operation is performed. In accordance with one feature which is used in some embodiments, a pallet position sensor **305** is used, in step **1024**, to determine if, as a result of the pickup operation performed in step **1022**, successful nesting of the pallet against the backrest **321** was achieved. Successful nesting is important because successful nesting decreases the chance that the pallet will slip off the forks or shift as the robotic device **300**, **600** or **800** moves from the pickup location to the target location where the pallet is to be placed.

[0115] If in step **1024** it is determined that the pallet position sensor **305** determined that full nesting was achieved, operation proceeds to step **1026**, in which the pallet is transported to the destination location, e.g., by the robotic device driving to the location with the pallet nested on its forks and in the air.

[0116] Unfortunately, due to imprecise positioning of the robotic forklift **300**, **600** or **800**, pallet nesting may not be achieved in some cases by the initial pallet pickup. In such a case the process will proceed from step **1024** to step **1016** to complete the pickup using a double pick operation with the initial pickup **1022** serving as the first pickup of the double pick procedure.

[0117] In step **1016**, the pallet which was picked up in step **1022** or step **1014**, but which did not end up fully against the forklift backrest **321**, i.e., did not achieve full nesting, is moved closer to the forklift, e.g., by retracting the forks while the pallet is in the air on the forks. Operation proceeds from step **1016** to step **1018**, in which the pallet is deposited on the ground or rack from which it was picked but this time at a position closer to the robotic device **300**, **600**, **800** than during the initial pick operation. This repositioning of the pallet closer to the forklift for the second pick operation performed as part of the double pick procedure increases the chance that full nesting will be achieved during the second pick operation. Operation proceeds from step **1018** to step **1020**, in which the pallet is picked up again. Pickup **1020** is a second or subsequent pickup operation which gives rise to the phase “double pick” operation.

[0118] Operation proceeds from step **1020** to step **1024** in which successful nesting of the pallet against the backrest is once again checked by sensor **305**. If successful nesting was not achieved, operation proceeds once again to step **1016** so that another attempt and moving and repicking the pallet can be implemented in an attempt to achieve full nesting. However, if as expected the pickup operation in step **1020** achieves full nesting of the pallet against the backrest **321**, operation will proceed to transport step **1026** with the pallet being moved to the intended destination.

[0119] By using the output of a pallet position sensor **305** to confirm proper nesting of a pallet prior to the robotic forklift **300**, **600** or **800** moving to a new location with the pallet in the air, safety is enhanced as compared to systems where proper pallet nesting against the fork assembly backrest **321** is not checked before the forklift moves the pallet.

[0120] From the above discussion it should be appreciated that the robotic forklift device **300**, **600** or **800** can be automatically maneuvered to a pickup location (P). Normally the pickup location P is sufficiently close to the pallet that a pallet pickup can be implemented by extending the forks into the pallet until nesting is achieved and then lifting the pallet into the air after which the forklift may move to a new position. As discussed above, the output of the pallet position sensor **305**, which is used in some embodiments, confirms and/or determines if nesting has been achieved prior to the forklift moving from the pickup location P.

[0121] While in many cases the forklift is positioned so that the pallet can be nested using a single fork extension operation, in some cases due to forklift miss-positioning, e.g., due to position sensor inaccuracy, nesting might not be achieved. In the case where a pallet is not nested, it rests on a portion of the fork with a gap between the pallet and back of the fork backrest **321**. The further the pallet is out from the rear of the forks/fork backrest **321**, the further away the center of gravity is from the main body of the forklift **300**, **600** or **800**. This can increase the chance of tipping or other issues. In addition, when not fully nested the pallet is more likely to slide since the area of contact, and thus the amount of friction between the forks and the pallet will often be less than what is achieved when the pallet is fully nested. Accordingly for safety reasons pallet nesting can be important.

[0122] The addition of the distance sensor **305** to the front of the forklift **300**, **600** or **800** which can sense the position of a pallet relative to the rear of the forks and/or fork backrest **321** allows for determining if nesting has been achieved.

[0123] If nesting has not been achieved a double bite procedure is implemented as discussed above. In a double bite procedure, the pallet is raised and moved in the direction of the forklift **300**, **600**, **800**, e.g., by retracting the extendable forks. The pallet is then lowered to the ground and the forks are once again extended until nesting is achieved. Once nesting is achieved the pallet is raised and the forklift moves to the intended destination.

[0124] A double bite procedure is also useful when, because of an obstruction such as a raised curb, pallet rack support or overhang, the forklift can not be positioned close enough to achieve pallet nesting in a single fork extension operation. In such a case, the forklift can be positioned as close to the pallet as reasonably possible. A lift operation is implemented with the pallet not being fully nested, and the pallet is moved closer to the fork mast by retracting the extended forks. The pallet is then placed on the ground, and the forks are once again extended, with nesting being confirmed before the forklift is moved from the pick location.

[0125] The use of the position sensor **305** allows for reliable confirmation of nesting of the pallet regardless of whether a normal or double bite pick operation is implemented. By using the output of the sensor **305** to confirm nesting and to trigger a double bite procedure when nesting is not achieved with a single fork placement, e.g., extension operation, safety and reliability is enhanced as compared to systems which do not include a distance sensor capable of sensing the pallet position at the front of the forklift assembly.

[0126] The implementation of a single or double pick operation will be explained now in further detail with reference to FIGS. **11-23** which show the exemplary robotic material handling device **300** in various positions and being used to perform various operations. Reference numbers in FIGS. **11-23** which are the same refer to the same or similar element and thus will not be described with respect to each figure for the sake of brevity.

[0127] FIG. **11** is a diagram **1100** showing the exemplary robotic device **300**, e.g., robotic forklift on flat ground **1102**. Fork distance D **1106** associated with the robotic forklift corresponds to the length of the forks. P **1104** is used to indicate the parking location, e.g., the location of the robotic device **300** main body from which the forks can be extended, raised, lower and/or titled (e.g., by tilting the mast to which the fork assembly is attached in some embodiments).

[0128] FIG. **12** is a diagram **1200** showing how the forks of the robotic device **300** of FIG. **11** can be extended out and away from the main body of the forklift **300**, e.g., extending the scissor like fork extension mechanism **397**. By extending the extension mechanism **397** the forks having a length D **1106** are extended out an extension distance E **1202** with the distance D+E **1108** representing the combination of extension distance E and fork length distance D.

[0129] FIG. **13** is a diagram **1300** which shows the robotic forklift **300** of FIG. **11** on level ground **1102** in a position where it is close to a pallet **1314** and preparing to initiate a pallet pick up operation to lift and move the pallet **1314** with the goods **1316**.

[0130] FIG. **14** is a diagram **1400** showing the robotic forklift **300** of FIG. **11** on level ground **1102** with the forks extended as part of a pallet pick up operation with the pallet **1314** being nested in the forks and ready for lifting from the ground **1102**. Note that full nesting has been achieved with no gap being visible between the pallet **1314** and fork backrest **321**.

[0131] FIGS. **13** and **14** shows an example of operation on level ground **1102** where the forklift **300** can come close to the pallet **1314** to be picked. Thus FIGS. **13** and **14** are an example where the pickup operation can be implemented without the need for a double pick operation but where the pallet position sensor **305** still being used to confirm full nesting of the pallet **1314** against the backrest **321** prior to the robotic device **300** moving with the pallet to a new location.

[0132] FIGS. **15-23** show operation of the robotic forklift **300** in an area in which the forklift's ability to approach the pallet **1314** to be picked up and moved is obstructed, e.g., by a step. The ground **1502** on which the forklift **300** operates is below the step **1504** on which the pallet **1314** is located and limits where the forklift **300** can park to implement the pickup operation. The step is merely exemplary of an obstruction that might limit forklift placement. Obstacles, rack overhangs, and/or pallets extending out from a multi-level storage rack might also limit the ability to position the forklift **300** close to the pallet to be picked up.

[0133] In FIG. **15** drawing **1500** shows the pallet **1314** to be picked up close to the edge of step **1504** and in a position allowing for a pickup from the step **1504** in a single pick operation.

[0134] In contrast to the FIG. **15** example, in the drawing **1600** of FIG. **16**, the pallet **1314** is shown set back from the edge of step **1504** making it difficult for the forklift to reach, pickup and achieve nesting of the pallet **1314** in a single pickup operation. FIG. **16** shows an example where a decision may be made to perform a double pick procedure or the double pick procedure may be triggered automatically by sensor **305** failing to achieve nesting after an initial pickup attempt.

[0135] FIG. **17** shows a diagram **1700** in which the robotic forklift **300** is performing the first step of a two bite (e.g., double bite) pickup procedure in which the pallet **1314** will be raised, moved, lowered and the forks reinserted to achieve proper pallet nesting before the robotic forklift **300** moves to take the pallet **1314**, and goods **1316** positioned thereon, to a new location. In FIG. **17** the forks are being extended into the pallet **1314**, but full nesting is not achieved as shown by the gap between the pallet **1314** and backrest **321**.

[0136] FIG. **18** shows the robotic forklift **300** lifting the pallet **1314** shown in FIG. **17** with the pickup fails to achieve nesting. The sensor **305** will detect the failure to achieve full nesting, e.g., in step **1024**, and operation will proceed based on the sensor failure to detect proper nesting.

[0137] FIG. **19** is a diagram **1900** showing the robotic forklift **300** retracting the pallet forks to move the pallet **1314** closer to the forklift **300** while the pallet **1314** is in the air so it can be lowered and repositioned as part of the double bite pickup procedure. FIG. **19** corresponds to step **1016** of FIG. **10**.

[0138] FIG. **20** is a diagram **2000** showing the robotic forklift **300** after it has lowered the pallet **1314** to the ground as part step **1018** of the double bite pickup procedure discussed with regard to FIG. **10**.

[0139] FIG. **21** is a diagram showing the forks of the robotic device **300** reinserted into the pallet **1314** to achieve full nesting of the pallet **1314** as part of the double bite pickup procedure. The reinsertion is part of the pallet pickup operation **1020** of FIG. **10**.

[0140] FIG. **22** is a diagram **2200** showing the pallet **1314** having been lifted following full nesting of the pallet **1314** as part of the double bite pickup procedure and now being ready for transport. As a result of successful pallet nesting, in step **1024** the processor controlling the robotic forklift **300** will determine based on the output of pallet position sensor **305** that the pallet is fully nested and can be safely transported to the intended pallet destination.

[0141] FIG. **23** is a diagram **2300** showing the pallet **1314** fully nested and being transported by the robotic forklift **300** to a new location away from the raised step **1504** where the pallet **1314** was previously stored. FIG. **23** corresponds to and represents the transport performed in step **1026** of FIG. **10**.

[0142] In various embodiments, a combination of sensors, e.g., a camera array **392** and/or other 3D sensors (e.g., radars **380**, **382**, **384** and/or Lidar units **386**, **388**, **390**), are used to determine information about the size, shape and/or location of, e.g., a pallet of goods to be moved and/or a location where the goods are to be placed. In various embodiments a camera array **392** and/or other sensors **380**, **382**, **384***m* **386**, **388**, **390** is used to capture information, e.g., images or measurements, of the goods and/or a location where the goods are to be placed. In some embodiments stereoscopic analysis of the captured images and/or analysis of the output of other sensors radars **380**, **382**, **384** and/or Lidar units **386**, **388**, **390**, provides 3D information about the pallet of goods and/or placement location.

[0143] A check is made to determine the 3D space, e.g., volume occupied by the goods being moved and outer dimensions of the volume. A similar analysis is made of the location, available storage space in a pallet rack, where the goods are to be placed. A check is made to determine if the goods will fit in the available space based on the 3D analysis of the goods and location where the goods are to be placed.

[0144] In some embodiments a first image and/or representation of the pallet of goods being moved is displayed to a material handling operator, e.g., a robotic forklift operator working at an operator workstation **200** from which the robotic device **300** can be manually controlled during periods of operation in which the device **300** is not operating autonomously.

[0145] In some embodiments, an image of a pallet with the goods being moved is displayed along side, in front of, or superimposed on an image or representation of the area where the goods are to be placed. Cuboids or other 3D shapes corresponding to the outside edges of the pallet and goods being moved are sometimes determined and used to indicate the volume which is expected to be occupied by the pallet and goods being moved and/or the volume available for storing the pallet of goods being moved. The 3D shape corresponding to the pallet of goods being moved, when displayed, helps an operator visually recognize the edges of the pallet and goods as well as the edges of the storage area along with potential impact areas. Areas of potential collision or impact are highlighted using colors or other visual indicia to help the operator focus his/her attention on areas of concern.

[0146] In various embodiments, an image of the pallet with goods, a 3D shape corresponding to the determined volume of the pallet with goods, a 3D shape corresponding to the available storage area, and/or an image of the storage area where a pallet with goods is to be placed is displayed to an operator, e.g., at an operator workstation **200** used to control the robotic device **300**.

[0147] Portions or areas of the pallet with goods, which are likely to suffer an impact during placement, are highlighted in some embodiments, as they are displayed, using coloration and/or shading to draw attention to areas of predicted impact or potential impact.

[0148] In some embodiments an image of the area where the goods are to be placed, is displayed alongside or behind the image of the goods to be moved, e.g., with the pallet of goods being superimposed on an image of the storage location in some embodiments. As noted above, in various embodiments the areas of potential impact are highlighted or otherwise identified to the operator. In this way a human operator is made aware of areas of concern. The operator can respond to indications of potential collisions with objects in the area where a pallet or other item is being moved by repositioning the pallet so that it will safely fit in the space or by choosing a new location to store the goods/pallet.

[0149] An exemplary method in which the 3D shape of a pallet with goods being moved is determined and checked against the available space at the location where the pallet of goods is to be placed will now be explained with reference to FIGS. **24**A, **24**B, **24**C and **24**D and FIG. **24**.

[0150] FIG. **24** shows a diagram **2400** showing how the diagrams **2401**, **2402**, **2403** and **2404** of FIGS. **24**A, **24**B, **24**C and **24**D can be combined to form a complete flow chart showing steps of an exemplary method.

[0151] The method starts in step **2410** of FIG. **24**A with a robotic device **300**, e.g., a forklift capable of operating in an autonomous mode or under direction of a human operator, e.g., working at an operator workstation **200**, being powered on and placed in a state in which it is ready to receive instructions, e.g., commands, and perform operations. Operation proceeds from start step **2410** to instruction receipt step **2412** in which the robotic device **300** receives instructions, e.g., in the form of a set of commands from a control system and/or operator workstation **200** which assigns operations, e.g., pallet pickup, transport, e.g., move, and place operations, to be performed to robotic device **300**.

[0152] FIG. **25** is a diagram **2500** illustrating an exemplary pallet **2502** with payload **2504**, a set of boxes, which can be picked up and moved in accordance with instructions received in step **2412**.

[0153] Operation proceeds from step **2412** to step **2114** in which the robotic device **300** moves to the pallet pickup location where it will position itself to pick up a pallet, e.g., with a payload such as a stack of boxes. The robotic device **300** may and sometimes does stop at the pallet pickup location, e.g., directly in front of the pallet **2502**, to capture images of the pallet **2502** with payload before the forks of the robotic device **300** are inserted into the pallet **2502**.

[0154] With the move to the pallet pickup location complete operation proceeds to step **2416** in which a first set of images of the pallet **2502** with payload **2504** is captured, e.g., the cameras **392**, **294**, **396** which are positioned at various locations on the robotic device **300**. Since the cameras are spaced apart from one another at known distances, stereoscopic processing of the images can be used to obtain 3D depth information about the pallet and payload. In addition to capturing images, in step **2416** other sensors, e.g. radar units **380**, **382 384** and/or LIDAR units **386**, **388**, **390** can be activated and used to capture depth information which can be used to generate a 3D representation of the pallet **2502** with payload **2504**.

[0155] With the first set of information sufficient to determine the 3D shape of the pallet **2502** with payload **2504** to be moved having been captured in step **2416**, e.g., by the cameras and/or sensors on the robotic device **300**, operation proceeds to step **2417** in which the captured information is communicated back to the operator workstation **200** for processing by the processor **202** and/to the processor **305** in the robotic device **300**.

[0156] Operation proceeds from step **2416** to store **2418** in which a first set of 3D information, e.g., a representation of the actual shape of the pallet **2502** and payload **2504** or a cuboid shape, e.g., such as the cuboid **2602** shown in FIG. **26**, corresponding to the outer perimeter of the pallet **2502** and payload are generated, e.g., by a processor which received the information captured in step **2416** at the operator workstation **200** or within the robotic device **300**. Step **2418** in some embodiments includes one or more of steps **2420** and **2422**. In step **2420** the first set of captured images is processed to generate a first 3D representation of the pallet **2502** with payload **2503**. Stereoscopic processing maybe and sometimes is used in step **2402** to generate the 3D representation of the pallet **2502** with payload **2504**. In other embodiments step **2420** processes captured images and combines it with lidar and/or radar data to generate the 3D representation while in other embodiments simply lidar or radar information is used to generate the 3D representation of the pallet and payload produced in step **2420**.

[0157] In step **2422** a 3D shape, e.g., a cuboid, corresponding to the perimeter of the pallet **2502** with payload **2504** is generated from the images and/or sensor data. Use of a cuboid shape rather than a detailed 3D model of the observed pallet and payload has the advantage of allowing for simplified comparisons of volumes when a cuboid is also used to represent the available space at the storage location where the pallet **2502** with payload **2504** is to be placed.

[0158] Referring now briefly to FIG. **26**, FIG. **26** is a diagram **2600** of the exemplary the pallet **2502** with payload **2504** of FIG. **25** with a 3D shape **2602**, e.g., a cuboid shape, corresponding to outer edges of the pallet with payload shown. The cuboid **2602** is the shape that was determined from captured images and/or other sensor date in step **2418**. The pallet **2502** with payload **2504** and corresponding shape **2602** is in some embodiments displayed on the display of the operator workstation **200** to an operator who controls pallet movement from the operator workstation **200**, The display of the captured image of the pallet and payload along with the cuboid shape superimposed thereon help the operator understand the volume which will be required at a storage location to store the pallet **2502** with goods **2504**.

[0159] With a 3D representation of the pallet **2502** with payload **2504**, before the transport of the pallet **2502** to the placement location, having been generated in step **2418**, operation proceeds to step **2424**. In step **2424** the pallet **2502** with payload **2504** is picked up and moved by the robotic device **300**, e.g., robotic forklift, to the intended pallet placement location indicated by the information received in step **2412**. The pickup and movement maybe and sometimes is implemented as an autonomous mode forklift operation but can be performed under operator control in some cases.

[0160] The robotic device **300** stops at the pallet placement location before moving the forks of the robotic device **300** to insert the pallet and payload into the intended storage location.

[0161] In some but not all embodiments a check on the payload is made to determine if it shifted while being moved. While this check is optional and involves capturing additional images and generating a second 3D representation or object corresponding to the pallet **2502** and payload **2504** it is useful in those cases where concern about goods shifting during movement of the pallet is an issue. The FIG. **24** example includes such a check with additional images and an additional set of 3D information being collected and used to generate a second 3D representation of the pallet **2502** and payload **2504** to determine if a shift in goods, e.g., a above a predetermined amount/distance, merits additional inspection and/or repacking of the payload **2504** on the pallet.

[0162] FIG. **27** represents the type of image(s) that may be captured after movement of the pallet **2502** with goods **2504**. Referring now briefly to FIG. **27**, FIG. **27** illustrates the exemplary pallet with payload of FIGS. **25** and **26** with some of the boxes **2702** on the pallet having shifted, e.g., to the right, during the move from the original pallet pickup location. Note that the set of 4 shifted boxes corresponding to reference number **2702** extends beyond and over the right side of the pallet **2502** with X **2704** representing the distance the boxes shifted to the right. As a result, the shift in boxes the volume required to store the pallet **2502** and payload **2504** is now larger than when it was originally picked up.

[0163] Referring once again to FIG. **24**A, if a check of whether the goods shifted is not to be made, steps **2426** to **2448** can be skipped with operation proceeding directly from step **2424** to step **2452**. However, in embodiments such as the one shown in FIG. **24** where a check is made to determine if the goods shifted while being moved, operation proceeds from step **2424** to step **2426** in which a second set of images and/or 3D sensor data of the pallet **2502** and payload **2504** is made. In step **2426** a second set of images of the pallet **2402** and payload **2504** is captured before operation proceeds to step **2428** which involves the same or similar operation to those performed in step **2418** but with the processing using the second set of images, e.g., images showing any shift.

[0164] In step **2428** after being communicated to the processor which is doing the processing, the images and/or other sensor data captured in step **2426** is processed to generate a second 3D representation, e.g., cuboid or model, corresponding to the pallet **2502** with goods **2504**. In some embodiments step **2428** includes one or both of steps **2430** and **2432**. In step **2430** the second set of captured images and/or other sensed data is processed to generate a second 3D representation of the pallet **2502** with payload **2504**. This representation will reflect the shift in goods **2702** that occurred when the pallet **2502** was moved from the pickup location to the intended storage location. In step **2458** a cuboid shape, such as the cuboid **2802** shown in FIG. **28** is generated corresponding to the perimeter of the pallet **2502** and goods **2504**.

[0165] Referring now briefly to FIG. **28**, it can be seen that FIG. **28** is a drawing **2800** that illustrates the exemplary the pallet **2502** with payload **2504** of FIG. **27** that shifted position during transport. The 3D shape **2802**, corresponding to outer edges of the pallet **2504** with payload, was determined in step **2428** and in step **2432** in particular. A captured image of the pallet **2502** with the payload **2504** showing the shifted goods **2702** with the cuboid **2802** is displayed in some embodiments to an operator using the operator workstation **200** to help the operator appreciate that shift in goods. Note that while the left, top, front and rear perimeter of the payload **2504** remains the same but the right side has now extended by the distance X as compared to in the FIGS. **25** and **26** examples. Thus, the cuboid **2802** used to represent the outer perimeter of the 3D volume which will be used or required to store the pallet **2502** and payload **2504** is now X inches larger on the right side than it was in the FIG. **26** example. FIG. **29** is a drawing **2900** showing an exemplary pallet rack **2902** which includes four storage locations **2904**, **2906**, **2908** and **2910** for storing goods. For purposes of explaining the invention the upper left storage location **2904** will be used as the initial intended storage location into which the pallet with goods **5404** is to moved with the upper right storage location **2908** being an alternative storage location that is also empty available for use.

[0166] The storage location **2904** may be and sometimes is initially selected as the location to which the pallet **2502** with goods is to be moved since the pallet **2502** can fit in the location if none of the goods on the pallet extend over the edge of the pallet **2502**.

[0167] Referring once again to FIG. **24**A, operation proceeds from step **2428** to step **2436** of FIG. **24**B via connecting node **2434**.

[0168] In step **2436** 3D information corresponding to the pallet **2502** with payload **2504** corresponding to the shape before and after the move is compared to determine the amount of shift, if any, in the goods **2504** on the pallet which occurred during the move. In various embodiments step **2436** includes one or both of steps **2438** and **2440**. In step **2438** the first and second 3D representations, e.g., models corresponding to before and after the move, of the pallet **2502** and payload **2504** are compared to identify any shift(s) in the payload. In step **2440** the 1st and 2nd cuboids shapes, e.g., the shapes corresponding to before and after the move, are compared to identify any shift(s) in the payload **2504**.

[0169] Based on the comparison(s) performed in step **2426**, as applied to the examples shown in FIGS. **26-28**, a shift of x inches to the right of at least some of the good **2504**, e.g., the boxes **2702**, will be determined with no other shift being detected.

[0170] With any shift of goods due to transportation between the pickup and intended placement site having been detected and measured in step **2436** operation proceeds to step **2442** in which a check is made to determine if a shift in payload which is likely to result in an unsuccessful move, e.g., failure of the pallet move due to goods falling off the pallet and/or a pallet placement is likely to be unsuccessful due to a shift in goods is made. In some embodiments step **2442** includes checking a detected amount of shift in the goods to determine if it exceeds a predetermined distance which is likely to result in a placement failure or goods falling off the pallet. This involves in some embodiments the shift distance X determined in step **2436** being compared to a threshold value, e.g., a set number of inches, which when exceeded is likely to result in a placement failure (e.g., due to the goods sticking too far over the edge of the pallet increasing the chance of a collision during placement and/or the goods falling off as the pallet is inserted into the storage location).

[0171] If in step **2442** it is determined that a shift in payload is likely to result in an unsuccessful placement or move, e.g., because in step **244** it is determined that a shift exceeds the predetermined maximum safe shift in goods, operation proceeds to step **2446** in which the robotic device **300** is controlled to transport the pallet **2502** with goods **2504** to an inspection and/or repacking area, e.g., by a command from the operator workstation **200** or processor of the robotic device **200**. Note that the shift in goods may create a risk of failure even if the shifted goods and pallet can still fit in the space available at the intended storage location. Accordingly, the check for a goods shift can be separate from and in addition to a later check as to whether the pallet **2502** and shifted goods will fit in the intended storage location **2904**.

[0172] After reaching the inspection area the goods **2504** are inspected, restacked and/or repackaged so that they are securely stored on the pallet **2502**. From step **2448** operation proceeds via connecting node **2450** to step **2416** where images of the pallet with payload are captured by cameras **392**, **294**, **396** and/or measurements by other sensors such as sonar/radar/lidar sensors **374**, **376**, **380**, **382**, **386**, **388**, **390** are captured before the repacked pallet **2502** is moved from the inspection/repacking area to the intended storage location.

[0173] Referring once again to step **2442** if a shift in payload which is likely to result in an unsuccess move and/or pallet placement is not detected, e.g., the goods did not shift during transport and/or it is determined in step **2444** that the amount of shift does not result in the goods extending over the edge of the pallet **2502** by the predetermined amount, operation proceeds to step **2452**. While the shift may not result in the goods extending over the edge of the pallet **2502** by the amount which would automatically trigger inspection/repacking, they may extend slightly over the edge of the pallet **2502** due to a shift of less than the predetermined distance that triggers an inspection.

[0174] In step **2452** a set of images of the pallet placement areas where the pallet **2502** with goods is to be placed is captured, e.g., using multiple cameras included in one or more of camera system **392**, **294**, **396** and/or other sensor data such as sonar, lidar and/or radar sensor data is captured, e.g., using sonar, radar and/or lidar units, to allow the shape and volume of the pallet placement area to be determined.

[0175] In the case of the FIG. **29** pallet rack **2902**, images and/or other sensor data are captured in step **2541** and used to generate 3D measurements of pallet storage areas **2904**, **2906**, **2908** and **2910**.

[0176] Operation proceeds from step **2452** to step **2453** in which the captured images/sensor data are communicated to a processor, e.g., the processor **202** of the operator workstation **200** or the processor **306** of the robotic device **300** for processing. The processor to which the captured images/sensor data is sent is normally the same processor to which was used to perform steps **2418** and/or step **2428** which involved generation of the 3D model or other 3D representation of the pallet **2502** with payload **2504**. With images and/or other sensor data having been captured in step **2452** operation proceeds to step **2454** in which 3D information, e.g., a 3D representation and/or cuboid corresponding to the intended pallet placement location is generated from the captured images and/or sensor data collected in step **2452**.

[0177] Thus, operation proceeds from step **2453** to step **2454** in which the processor to which the images/sensor data were communicated generates 3D information corresponding to the pallet placement location, e.g., a 3D representation of the available space for storing the pallet or a cuboid corresponding to the perimeter of the available space, e.g., the inside unobstructed volume of a slot in a pallet rack into which the pallet can be inserted. In some embodiments step **2454** includes one or both of steps **2456** and **2458**. In step **2456** the processor processing the set of images/sensor data corresponding to the intended pallet storage location generates a 3D representation of the pallet placement location, e.g., a 3D model of the available space. In step **2458** the processor determines a cuboid shape corresponding to the available space at the pallet placement location. A cuboid is a three-dimensional shape which is a convex polyhedron that is bounded by six rectangular faces with eight vertices and twelve edges. A cuboid is also sometimes called a rectangular prism. An example of a cuboid in real life is a rectangular box. The cuboid shape is, in essence, a simplified 3D representation of the available space since it has flat sides and is a reasonable representation of the area into which a pallet and payload can be inserted without the complexities of a more precise 3D model which may have curved or unusually shaped surfaces.

[0178] A cuboid can be easily determined as opposed to a more complex model of the available space and/or pallet with goods and is therefore used in some embodiments to represent the space required for the pallet with goods as well as the available space at a pallet storage location. Cuboids can also be easily compared to determine if one cuboid will fit inside another. This makes the use of cuboids for the pallet with goods as well as the storage location well suited for use in determining if sufficient space is available to store the pallet **2502** with goods **2504**.

[0179] FIG. **30** is a diagram **3000** showing the pallet rack **2902** with storage locations **2904** and **2908** and corresponding cuboids **3002**, **3004**, respectively which are determined in step **2454**. Cuboid **3002** is a 3D representation of the available storage space at storage location **2904** while cuboid **3004** is a 3D representation of the available storage space at storage location **2908**. Note that is a product or other obstruction were present in a location the cuboid corresponding to the storage location would be reduced in size to avoid the obstruction resulting in a much smaller cuboid than the one shown since the available space would be reduced by the presence of the object/obstruction. This may be the case when a box or other item fell off a pallet previously stored in the location and/or off a pallet located to the side or behind the storage location. The 3D check of the available space will detect the presence of such an unexpected object before the pallet is inserted thereby avoiding a collision or product damage.

[0180] With a 3D representation of the available space for pallet and payload placement having been generated in step **2454** operation proceeds via connecting node C **2460** to step **2462** shown in FIG. **24**C.

[0181] In step **2462** a check is made to determine if there is available space at the intended pallet placement location for the pallet **2502** with the payload **2504**. Step **2463** in some cases determining if there is a collision risk between the pallet and payload and pallet rack/previously stored goods and, if so, what portions of the pallet **2502** and payload **2504** are at risk of a collision during placement of the pallet and payload in the storage location so that the portions at risk can be, and in some embodiments are visually highlighted to an operator controlling robotic device operation such as pallet placement, e.g., with the operator workstation using red or other coloring to highlight/indicate areas at risk of a collision.

[0182] In some embodiments step **2462** includes one, more or all of steps **2464**, **2468**, **2470**, **2472**, **2474** which are implemented by the processor **202** in the operator workstation **200** and/or the processor **306** in the robotic device **300**. In step **2464** which in some embodiments is part of step **2462**, the 3D shape **2802** corresponding to the pallet **2502** with payload **2504** is compared to the determined available space/volume for pallet storage at the storage location where the pallet is to be placed, e.g., which in the example shown in FIG. **31** is represented by the 3D shape **3002** included in the diagram **3100**. This is to determine if the pallet **2502** with goods **2504** will fit in the available storage space where the pallet with goods is to be placed. In some embodiments step **2464** includes step **2466** in which a cuboid shape **2802** corresponding to the pallet **2502** with goods **2504** is compared to a cuboid shape **3002** corresponding to the available storage space. The cuboid **2802** corresponding to the pallet **2502** with payload **2504** should be smaller on all sides than the cuboid **3003** corresponding to the available storage space, e.g., by a predetermined safety margin in terms of size, e.g., 6 inches or a foot for example, with the cuboid for the pallet **2502** with goods **2504** required in some embodiments to be at least 1 foot smaller on all sides than the cuboid corresponding to the available storage space. A distance of less than the predetermined safety margin indicates a possible or likely collision on a given side. While 1 foot safety margin is used as an example smaller or larger safety margins are possible, e.g., depending on how controllable the robotic device is and the level of precision with which placement operations can be made.

[0183] As a result of the comparisons made in step **2464**, areas, e.g., sides, where collisions might occur and the probability of a collision is determined, e.g., with higher probability of a collusion the smaller the space difference between the side of the pallet **2502** with payload **2504** and the side the storage area into which it is inserted. When there is no predicted space between the side of the pallet **2502** with payload **2504**, e.g., because the pallet with payload is larger on one side than the space available on that side, a collision is predicted.

[0184] Operation proceeds from step **2464** to step **2468** in which visual indicators, different colors, are assigned to different portions, e.g., sides/top/bottom, of the 3D shapes, corresponding to the pallet **2502** with payload **2504**. The visual indicator, e.g., color indicates the risk of collision. For example, in some embodiments red is used to indicate an expected collision, green to indicate a safe area and orange to indicate an area where a collision might occur if the pallet is not carefully moved in position in the storage location. The visual indications can be and sometimes are displayed to an operator controlling the movement of the robotic device by highlighting and coloring portions of cuboids on a display screen presented to the operator at the operator workstation **300**.

[0185] FIG. **32** is a diagram **3200** which includes an image of the pallet rack **2902**, pallet **2502** with goods **2504** and cuboids **3002**, **2802** with the right side **3104** of the cuboid **2802** being highlighted/indicated in red, represented by the dark black lines forming the right side edges of the cuboid **2802**. The visual indication **3104** is intended to visual indicate to an operator of an operator workstation on which the drawing **3200** is displayed that a collision is predicted on the right side of the pallet **2502** with payload **2504** if it is inserted into the top left pallet storage location. This predicted collision is with pallet post **3102** and is due to the shift of goods, i.e., the top four right side boxes, when the pallet **2502** was moved.

[0186] Operation proceeds from step **2468** to step **2470** in which an automatic check is made by the processor of the operator workstation **200** or robotic device **300** to determine if a collision will happen or expected to happen if the pallet is inserted into the intended storage location. This check may and sometimes does involve checking if there is at least the predetermined safety margin of distance, e.g., 1 foot, available on all sides of the pallet and payload at the storage location. If there is no safety margin or a space less than the predetermined safety margin on any side or the top, a decision is made in step **2470** that a collision will happen (e.g., in the case of no safety margin on a side) or is likely to happen (in the case of a distance less than the predetermined safety margin being available on one side or top of the pallet **2502** and payload **2504**.

[0187] If in step **2470** it is determined that a collision will occur or is likely to occur, operation proceeds via the Y decision path to step **2474** in which the processor performing the check determines that sufficient space is not available for successful pallet placement. However, if in step **2470** it is determined that a collision will not happen or is unlikely to happen, e.g., because there is at least the predetermined safety margin in terms of space on all sides and the top of the pallet **2502** with payload as compared to the storage location, operation proceeds from step **2470** via N path and in step **2472** it is determined that there is sufficient space available for the placement of the pallet **2502** with payload **2504**.

[0188] Operation proceeds from step **2472** (placement success predicted) and **2474** (placement predicted to fail) to step **2478** of FIG. **24**D via connecting node D **2476**.

[0189] In step **2478** an image, e.g., an image such as diagram **3200** shown in FIG. **3200**, is displayed to the operator working at the operator workstation **200** used to control the robotic device to provide the operator with information about the space available for pallet placement and the size of the pallet along with areas where collisions are possible or expected. The display operation can take many forms with the operator being presented with the 3D shapes (cuboids) corresponding to the pallet **2502** with payload **2504** and/or the storage location where the pallet is to be placed along with an image of the pallet and/or image of the storage location. Such images and shapes can be and sometimes are superimposed on each other and/or displayed side by side or one in front of the other to facilitate a spatial understanding of how the pallet **2502** with payload **2504** needs to be manipulated/moved to store it in the intended storage location, e.g., in an available space of a pallet rack. The display operation of step **2478** will occur when a failure is predicted and operator oversight/control is to be implemented. However, when successful placement is predicted, display step **2478** is optional and in some embodiments the placement of the pallet is performed autonomously without operator involvement. In such a case step **2478** can and sometimes is skipped since an operator need not be involved in the placement operation or control of the robotic device.

[0190] Step **2478** in some embodiments includes one, more than one, or all of steps **2480**, **2482**, **2484** and **2486**. In step **2486** an image is displayed to the operator of the operator workstation **200** of the pallet **2502** with payload **2504** along with the cuboid corresponding to the 3D shape of the pallet and goods. In step **2882** areas of collision concern, areas where an actual or possible collision are expected, are highlighted or otherwise indicated visually, e.g., using different colors to indicate different levels of risk of collision such as red for predicted collision, orange for possible collision and green for no collision expected. The colors may be and sometimes are applied to the sides of the cuboid corresponding to the pallet with payload to indicate the area of the pallet and/or payload at risk of collision and to draw the operators attention to such areas.

[0191] In step **2484** an image showing the cuboid corresponding to the pallet **2502** with payload **2504**, is displayed along with an image of the pallet **2502** with goods, as well as the cuboid corresponding to the storage location showing the space available for storing the pallet and goods. Thus, in such an embodiment, both the cuboid corresponding to the pallet **2502** with goods and the cuboid corresponding to the storage location are shown. The images may be side by side or with the cuboid corresponding to the pallet with goods in front of the cuboid corresponding to the storage location so that an operator can visually see how the pallet with goods can fit inside the storage location represented by the storage location size cuboid.

[0192] In step **2486** the cuboids corresponding to the pallet with goods and the storage location are highlighted using colors to indicate areas where a collision, e.g., the right side of the pallet with goods as indicated by reference **3104**, is predicted or likely to occur. The cuboids **3002**, **2802** can be and sometimes are superimposed on an image captured from the forklift showing the pallet **2502** with payload **2504** in front of the intended storage location with the cuboid **2802** corresponding to the pallet **2502** with goods **2504** being superimposed on the pallet **2502** with goods and the cuboid **3002** corresponding to the available space at storage location **2904** where the pallet **2502** is to be placed being superimposed on an image of the storage location as represented by the image of the rack **2902** in FIG. **32**. The sides/top of the cuboids which are not likely to collide in one embodiment are shown in green, sides/top which are predicted to possibly collide are shown in orange and sides/top where collision is predicted are shown in red. This in some embodiments on a single display screen shown to an operator both the pallet **2502** with goods **2504**, pallet rack with storage location and corresponding cuboids with color coding to show risk are visible facilitating operator control of the robotic device with the image being updated as the robotic device **300** moves the pallet to reflect the current relative positions of the various elements displayed to the operator.

[0193] Operation moves from step **2478** to step **2479** in which the robotic device **300** is controlled based on whether or not it was determined that there is sufficient space at the intended pallet storage location for successful pallet placement, e.g., with this determination having been made in step **2462**. In some embodiments if the it was determined that there was sufficient space available for successful pallet with payload placement, the robotic device **300** is controlled to autonomously proceed with pallet placement while if it was determined that there was insufficient space for successful pallet placement, e.g., a failure or collision was likely due to an insufficient space safety margin around the pallet **2502** with goods, operator assistance in controlling placement of the pallet is sought or another placement location having more available space is identified and used, e.g., with autonomous control of the robotic device during pallet placement at the new location. While the robotic device **300** is operating in autonomous mode, a human operator is not involved in controlling the robotic device. Thus, automatic selection of an alternative storage location can be efficient from the perspective of operator time not being required.

[0194] Step **2478** in some embodiments beings with step **2488** where a check is made as to whether or not sufficient space was determined to be available at the intended storage location for pallet placement, e.g., automatic pallet placement success or failure was predicted. If sufficient space was not determined to be present, pallet placement failure was predicted and operation proceeds from step **2488** along the no (N) path to step **2494**. If in step **2488** it was determined that sufficient space was available for successful pallet placement, operation proceeds from step **2488** along the yes (Y) path to step **2490** in which pallet placement in the storage location automatically proceeds. With the pallet **2502** with goods **2504** having been stored in step **2490**, operation proceeds via connecting node F **2492** back to step **2412**.

[0195] In step **2488** when a failure is predicted, operation proceeds to step **2494** in which the operator is notified of the predicted failure and provided an opportunity to manually control pallet placement, e.g., with the image generated and displayed in step **2478** being intended to help the operator control the pallet placement. In step **2495** a check is made as to whether the operator of the operator workstation **200** decided to proceed with pallet placement under manual control as may be indicated by the operator signaling the robotic device **300** on how to move the pallet **2502** into the intended storage location. If in step **2495** it is determined that the operator will manually control the robotic device **300** to place the pallet **2502** with goods **2504** into the storage location, operation proceeds via the YES path indicated by Y to step **2496** in which the robotic device **300** proceeds to place the pallet **2502** with goods **2504** into the intended storage location under manual operator control. With the pallet having been placed into the intended storage location in step **2496** under manual operator control, operation proceeds from step **2946** via connecting node F **2492** to step **2412** in which the robotic device is provided with instructions to implement another pallet move.

[0196] However, if in step **2495** it is determined that the operator has decided not to manually control pallet placement, e.g., as may be indicated by a signal from the operator to find a new storage location, operation proceeds from step **2470** to step **2497** in which the processor of the operator workstation **200** or robotic device **300** identifies, e.g., finds and selects, an alternative storage location with sufficient space for the pallet **2502** with payload **2504**. The identified alternative storage location is set in step **2494** to be used as the pallet placement location and operation proceeds via connecting node D **2499** To step **2424** which involves moving the pallet to the intended pallet placement location which in this case will be the alternative location that was selected in step **2497**. The movement is followed by the pallet placement related steps which were previously discussed. In such an embodiment where an alternative storage location, e.g., storage location **2908**, is selected due to insufficient space at the originally intended storage location and then the robotic device **300** can operate autonomously to move to the newly selected location and store the pallet **2502** with goods **2504**, e.g., after automatic checking of the new storage location for sufficient space for pallet storage.

[0197] By checking both the pallet **2502** with payload **2504** and the storage area into which the pallet is to be inserted for space, collisions due to shifting of goods during transport or unexpected objects at the storage location where the pallet is to be placed can be detected and collisions avoided.

[0198] FIG. **33** is a diagram **3300** showing the cuboid **3004** corresponding to storage location **2908** in the top right of the pallet rack **1902** being compared to the cuboid **2802** to make sure the goods will fit in the storage location **2908**. The comparison shows that there is space of at least the predetermined amount, e.g., 1 foot in some embodiments, which will avoid a collision on all sides and the top. Accordingly, it is determined that it is safe to automatically place the pallet **2502** and goods **2504** in the alternative storage location **2908**. In the example the robotic device **300** will autonomously or under the control of the operator complete the placement of the pallet **2502** with goods **2504** in the storage location **2908** since the space availability check made before insertion of the pallet passed the safety check.

[0199] FIG. **34** is a diagram **3400** showing the pallet **2502** with goods **2504** stored, e.g., automatically, after the storage location **2908** is selected as an alternative storage location due to a predicted collision if storage location **2904** was used. As can be seen the pallet **2502** with goods **2504** fits within the storage location **2908** and the collision that would have occurred if the pallet **2502** with shifted goods was inserted into storage location **2904** was avoided.

[0200] Numerous variations on the 3D dimensioning, checking and information display are possible while still providing the benefits obtained by the dimensioning and checking operations discussed above.

[0201] In the following sets of numbered exemplary embodiments, a reference to a previous numbered embodiment refers to a numbered embodiment in the same set.

First Set of Exemplary Numbered Method Embodiments

[0202] Method Embodiment 1. A method of operating a robotic material handling device (**300**, **600** or **800**) having a string of lights (e.g., light bar **307**, **607** or **807**) mounted thereon, comprising: collecting (**504**) sensor data; analyzing (**506**) the sensor data to detect obstacles; and controlling (**520**) a group of lights, in the string of lights, corresponding to a direction of a detected obstacle to be illuminated (e.g., light up) to indicate the detection of the detected obstacle.
[0203] Method Embodiment 2. The method of Method Embodiment 1, further comprising: determining (**514**) the direction of the detected obstacle relative to the robotic material handling device.
[0204] Method Embodiment 3. The method of Method Embodiment 2, further comprising: determining (**510**) a distance to the detected obstacle; and determining (**512**) a number of lights (e.g., the number of adjacent lights facing in the direction of the obstacle which form the group of lights that is illuminated) to be illuminated due to the detection of the obstacle based on the determined distance to the obstacle.
[0205] Method Embodiment 3A. The method of Method Embodiment 3, wherein the determined number of lights is greater when the detected distance is a first distance than when the detected distance is a second distance, said first distance being smaller than the second distance (e.g., the closer the detected object is the larger the number of lights which are illuminated to indicate detection of the object).
[0206] Method Embodiment 4. The method of Method Embodiment 3, further comprising: determining the type of obstacle which was detected (e.g., human or inanimate object).
[0207] Method Embodiment 5. The method of Method Embodiment 4, further comprising: determining (**518**) a color to be used for at least one of the lights in the group of lights to indicate the type of object which was detected.
[0208] Method Embodiment 6. The method of Method Embodiment 5, wherein controlling (**520**) the group of lights, in the string of lights, corresponding to the direction of the detected obstacle to be illuminated includes: controlling at least one light in the group of lights to be illuminated in the determined color.
[0209] Method Embodiment 7. The method of Method Embodiment 6, where the determined color is a first color (e.g., red) when the detected object is determined to be a human and a second color (e.g., orange) when the determined object is determined to be a non-human object.
[0210] Method Embodiment 8. The method of Method Embodiment 3, wherein said string of lights includes lights mounted on of a front, rear, left and right side of the robotic material handling vehicle.
[0211] Method Embodiment 9. The method of Method Embodiment 3, wherein said robotic material handling vehicle is a robotic forklift capable of operating without a human operator present in the robotic material handling vehicle, said string of lights being an obstacle detected indicator light bar (**307**).
[0212] Method Embodiment 10. The method of Method Embodiment 9, wherein said obstacle detected indicator light bar (**307**) extends around a canopy of said robotic material handling vehicle.
[0213] Method Embodiment 10A. The method of Method Embodiment 9, wherein said obstacle detected indicator light bar (**307** or **807** which includes light bar portions **807**′, **807**″, **807**‴ and **807**⁗) extends around a main body portion (**802**) of the robotic material handling vehicle.
First Set of Exemplary Numbered Apparatus Embodiments
[0214] Apparatus Embodiment 1. A robotic material handling device (**300**, **600** or **800**) comprising: one or more sensors (**370**, **372**, **374**, **376**, **392**, **394**); [0215] a string of lights (e.g., light bar **307**, **607** or **807**); and a processor (**399**) configured to: collect (**504**) sensor data; analyze (**506**) the sensor data to detect obstacles; and control (**520**) a group of lights, in the string of lights, corresponding to a direction of a detected obstacle to be illuminated (e.g., light up) to indicate the detection of the detected obstacle.

[0216] Apparatus Embodiment 2. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 1, wherein the processor (**399**) is further configured to: [0217] determine (**514**) the direction of the detected obstacle relative to the robotic material handling device.
[0218] Apparatus Embodiment 3. The robotic material handling device (**300**, **600** or **800**) method of Apparatus Embodiment 2, wherein the processor (**399**) is further configured to: determine (**510**) a distance to the detected obstacle; and determine (**512**) a number of lights (e.g., the number of adjacent lights facing in the direction of the obstacle which form the group of lights that is illuminated) to be illuminated due to the detection of the obstacle based on the determined distance to the obstacle.
[0219] Apparatus Embodiment 3A. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 3 wherein the determined number of lights is greater when the detected distance is a first distance than when the detected distance is a second distance, said first distance being smaller than the second distance (e.g., the closer the detected object is the larger the number of lights which are illuminated to indicate detection of the object).
[0220] Apparatus Embodiment 4. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 3, wherein the processor (**399**) is further configured to: determine the type of obstacle which was detected (e.g., human or inanimate object).
[0221] Apparatus Embodiment 5. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 4, wherein the processor is further configured to: determine (**518**) a color to be used for at least one of the lights in the group of lights to indicate the type of object which was detected.
[0222] Apparatus Embodiment 6. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 5, wherein controlling (**520**) the group of lights, in the string of lights, corresponding to the direction of the detected obstacle to be illuminated includes: controlling at least one light in the group of lights to be illuminated in the determined color.
[0223] Apparatus Embodiment 7. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 6, where the determined color is a first color (e.g., red) when the detected object is determined to be a human and a second color (e.g., orange) when the determined object is determined to be a non-human object.
[0224] Apparatus Embodiment 8. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 3, wherein said string of lights includes lights mounted on of a front, rear, left and right side of the robotic material handling vehicle.
[0225] Apparatus Embodiment 9. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 3, wherein said robotic material handling vehicle is a robotic forklift capable of operating without a human operator present in the robotic material handling vehicle, said string of lights being an obstacle detected indicator light bar (**307**).
[0226] Apparatus Embodiment 10. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 9, wherein said obstacle detected indicator light bar (**307** or **607** including light string portions **607**′, **607**″, **607**‴ and **607**″″) extends around a canopy **620** of said robotic material handling vehicle.
[0227] Apparatus Embodiment 10A. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 9, wherein said obstacle detected indicator light bar (**307** or **807** which includes light bar portions **807**′, **807**″, **807**‴ and **807**″″) extends around a main body portion (**802**) of the robotic material handling vehicle.
First Set of Exemplary Numbered Non-Transitory
Computer Readable Medium Embodiments
[0228] Non-Transitory Computer Readable Medium Embodiment 1. A non-transitory computer readable medium including processor executable instruction which when executed by a processor (**399**) of a robotic material handling device (**300**, **600** or **800**) having a string of lights (e.g., light bar **307**, **607** or **807**) mounted thereon, controls the robotic material handling vehicle to: collect

(**504**) sensor data; analyze (**506**) the sensor data to detect obstacles; and control (**520**) a group of lights, in the string of lights, corresponding to a direction of a detected obstacle to be illuminated (e.g., light up) to indicate the detection of the detected obstacle.

Second Set of Exemplary Numbered Method Embodiments

[0229] Method Embodiment 1. A method of controlling a robotic material handling device (**300**, **600** or **800**) comprising: controlling the robotic material handling device (**300**, **600** or **800**) to perform a first pallet pickup operation (**1022**) to pick up a pallet (**1314**); and determining (**1024**) from the output of a pallet position sensor (**305**) on the robotic material handling device (**300**, **600** or **800**) if the pallet (**1314**) is nested on forks (**323**, **325**) of the robotic material handling device (**300**, **600** or **800**).

[0230] Method Embodiment 2. The method of Method Embodiment 1, further comprising: controlling the robotic material handling device, in response to determining (**1024**) from the output of the pallet position sensor (**305**) that the pallet (**1314**) is not nested on the forks (**323**, **325**) (e.g., no output of step **1024**), to move (**1016**) the pallet (**1314**) closer to the robotic material handling device while the pallet (**1314**) is in the air on the forks (**323**, **325**) due to the first pallet pickup; controlling the robotic material handling device to deposit (**1018**) the pallet (**1314**) (e.g., on the ground or surface from which the pallet was first picked up but now at a position closer to the robotic material handling device); and controlling the robotic material handling device (**300**, **600** or **800**) to perform a second pallet pickup operation (**1020**) to pick up the pallet (**1314**) for a second time.

[0231] Method Embodiment 3. The method of Method Embodiment 2, further comprising: determining (**1024**), after the second pallet pickup operation (**1020**), from the output of the pallet position sensor (**305**) on the robotic material handling device (**300**, **600** or **800**) if the pallet (**1314**) is nested on forks (**323**, **325**) of the robotic material handling device (**300**, **600** or **800**).

[0232] Method Embodiment 4. The method of Method Embodiment 3, further comprising: in response to determining (**1024**), after the second pallet pickup operation (**1020**), that the pallet (**1314**) is nested on forks (**323**, **325**), transporting (**1026**) the pallet (**1314**) to a pallet destination.

[0233] Method Embodiment 5. The method of Method Embodiment 1, wherein said pallet position sensor (**305**) is mounted at a fixed position on a fork backrest that will move with the forks as the forks are extended or retracted (e.g., at a fixed position relative to a front surface of a vertical backrest member of the backrest which will contact a pallet when the pallet is fully nested on the forks), said fixed position being constant in some embodiments relative to the location of the tips of the pallet forks allowing for accurate measurement of how far the pallet is inserted into the forks based on the distance measured between the sensor and the pallet.

[0234] Method Embodiment 6. The method of Method Embodiment 5, wherein said pallet position sensor is a non-contact distance measuring device (e.g., an ultrasonic distance measuring device in some embodiments).

[0235] Method Embodiment 7. The method of Method Embodiment 6, wherein said pallet position sensor is mounted on a member of the fork backrest positioned at a location which is between the forks.

[0236] Method Embodiment 8. The method of Method Embodiment 7 wherein the pallet position sensor (**305**) is mounted above a forward looking camera array **392** which has a field of view including the pallet forks.

[0237] Method Embodiment 9. The method of Method Embodiment 4, further comprising: receiving (**1004**) an instruction to move the pallet (**1314**) to the pallet destination; determining (**1006**) the location of the pallet (**1314**) to be moved; determining (**1008**) a safe forklift placement location from which to pick the pallet to be moved; controlling the robotic material handling device to move (**1010**) to the determined forklift placement location prior to proceeding with lifting of the pallet (**1314**) (with in some cases the first and second pallet lifts being performed while the robotic material handling device remains stationary (e.g., parked) at the forklift placement position).

[0238] Method Embodiment 10. The method of Method Embodiment 1, wherein the robotic material handling device (**300**, **600** or **800**) is a robotic forklift capable of operating in a fully autonomous mode in which the processor (**399**) controls the robotic device to perform the recited steps.

Second Set of Exemplary Numbered Apparatus Embodiments

[0239] Apparatus Embodiment 1. A robotic material handling device (**300**, **600** or **800**) comprising: a fork assembly (**379**) including a backrest **321**, a first fork **323** and a second fork **325**; a pallet position sensor **305**; and a processor (**399**) coupled to the pallet position sensor (**305**) configured to control the robotic material handling device (**300**, **600** or **800**) to: perform a first pallet pickup operation (**1022**) to pick up a pallet (**1314**); and determine (**1024**) from the output of a pallet position sensor (**305**) on the robotic material handling device (**300**, **600** or **800**) if the pallet (**1314**) is nested on forks (**323**, **325**) of the robotic material handling device (**300**, **600** or **800**).

[0240] Apparatus Embodiment 2. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 1, wherein the processor (**399**) is further configured to control the robotic material handling device (**300**, **600** or **800**) to: move (**1016**) the pallet (**1314**) closer to the robotic material handling device while the pallet (**1314**) is in the air on the forks (**323**, **325**) due to the first pallet pickup, in response to determining (**1024**) from the output of the pallet position sensor (**305**) that the pallet (**1314**) is not nested on the forks (**323**, **325**) (e.g., no output of step **1024**); deposit (**1018**) the pallet (**1314**) (e.g., on the ground or surface from which the pallet was first picked up but now at a position closer to the robotic material handling device); and perform a second pallet pickup operation (**1020**) to pick up the pallet (**1314**) for a second time.

[0241] Apparatus Embodiment 3. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 2, further comprising: determining (**1024**), after the second pallet pickup operation (**1020**), from the output of the pallet position sensor (**305**) on the robotic material handling device (**300**, **600** or **800**) if the pallet (**1314**) is nested on forks (**323**, **325**) of the robotic material handling device (**300**, **600** or **800**).

[0242] Apparatus Embodiment 4. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 3, further comprising: in response to determining (**1024**), after the second pallet pickup operation (**1020**), that the pallet (**1314**) is nested on forks (**323**, **325**), transporting (**1026**) the pallet (**1314**) to a pallet destination.

[0243] Apparatus Embodiment 5. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 1, wherein said pallet position sensor (**305**) is mounted at a fixed position on a fork backrest that will move with the forks as the forks are extended or retracted (e.g., at a fixed position relative to a front surface of a vertical backrest member of the backrest which will contact a pallet when the pallet is fully nested on the forks), said fixed position being constant in some embodiments relative to the location of the tips of the pallet forks allowing for accurate measurement of how far the pallet is inserted into the forks based on the distance measured between the sensor and the pallet).

[0244] Apparatus Embodiment 6. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 5, wherein said pallet position sensor is a non-contact distance measuring device (e.g., an ultrasonic distance measuring device in some embodiments).

[0245] Apparatus Embodiment 7. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 6, wherein said pallet position sensor is mounted on a member of the fork backrest positioned at a location which is between the forks.

[0246] Apparatus Embodiment 8. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 7 wherein the pallet position sensor (**305**) is mounted above a forward looking camera array **392** which has a field of view including the pallet forks.

[0247] Apparatus Embodiment 9. The robotic material handling device (**300**, **600** or **800**) of Apparatus Embodiment 4, further comprising: receiving (**1004**) an instruction to move the pallet (**1314**) to the pallet destination; determining (**1006**) the location of the pallet (**1314**) to be moved;

determining (**1008**) a safe forklift placement location from which to pick the pallet to be moved; and controlling the robotic material handling device to move (**1010**) to the determined forklift placement location prior to proceeding with lifting of the pallet (**1314**) (with in some cases the first and second pallet lifts being performed while the robotic material handling device remains stationary (e.g., parked) at the forklift placement position).

Second Set of Numbered Non-transitory Computer Readable Medium Embodiments

[0248] Non-Transitory Computer Readable Medium Embodiment 1. A non-transitory computer readable medium which, when executed by a processor (e.g., processor **399**) in a robotic material handling device (**300**, **600** or **800**) control the robotic material handling device to: perform a first pallet pickup operation (**1022**) to pick up a pallet (**1314**); and determine (**1024**) from the output of a pallet position sensor (**305**) on the robotic material handling device (**300**, **600** or **800**) if the pallet (**1314**) is nested on forks (**323**, **325**) of the robotic material handling device (**300**, **600** or **800**).

Third Set of Exemplary Numbered Method Embodiments

[0249] Method Embodiment 1. A method of controlling a robotic device (**300**), the method comprising: controlling the robotic device to move (**2414**) to a pallet pickup location (e.g., under the control of instructions communicated by a control device such as operator workstation **200** and received by the robotic device in step **2412**); capturing (**2416**) (e.g., at the pallet pickup location) a first set of sensor data providing 3D information corresponding to a pallet **2502** with payload **2504** (e.g., images captured by cameras of a camera system (e.g., forward looking camera system **392**) which includes multiple cameras spaced apart from one another to support stereoscopic depth and/or shape determinations which are generated from images captured by the cameras of the camera system or LIDA, RADA or SONAR sensor data providing 3D information about the pallet and payload); and generating (**2418** 3D representation of pallet **2502** with payload **2504** before moving or **2428** 3D representation of pallet with payload after move) a 3D representation of the pallet **2502** with payload **2504** from captured sensor data.

[0250] Method Embodiment 2. The method of Method Embodiment 1, further comprising: capturing (**2452**) (e.g., at a location in front of a storage rack into which the pallet **2502** with payload **2504** is inserted) a set of sensor data providing 3D information corresponding to a pallet storage location; (e.g., images captured by cameras of the camera system (e.g., forward looking camera system **392**) which includes multiple cameras spaced apart from one another to support stereoscopic depth and/or shape determinations which are generated from images captured by the cameras of the camera system or LIDA, RADA or SONAR sensor data providing 3D information about the pallet and payload); generating (**2454**) a 3D representation of the available storage space at the pallet placement location from captured sensor data; and determining (**2462**) if there is sufficient space at the pallet placement location for successful pallet placement (e.g., determine if the robotic device **300** is likely to succeed or will succeed with placement of the pallet **2502** with goods **2504**, e.g., autonomously without operator assistance), said determining including comparing (**2464**) the generated 3D shape corresponding to the pallet **2502** with the payload **2504** to the generated 3D shape corresponding to the available storage space at the pallet placement location.

[0251] Method Embodiment 3. The method of Method Embodiment 2, further comprising: controlling (**2478**) robotic device (**300**) operation based on the determination (**2462**) of whether or not there is sufficient space at the pallet placement location for successful pallet placement.

[0252] Method Embodiment 4. The method of Method Embodiment 3, wherein capturing (**2416**) a first set of sensor data providing 3D information corresponding to a pallet **2502** with payload **2504** includes capturing a first set of images using cameras of a stereoscopic camera array (**392**); and wherein generating a 3D representation of the pallet **2502** with payload **2504** from captured sensor data includes processing the captured images to generate a 3D shape or model corresponding toe the pallet with payload.

[0253] Method Embodiment 5. The method of Method Embodiment 4, wherein generating (**2418**) a 3D representation of the pallet **2502** with payload **2504** from captured sensor data includes processing the captured images to generate (**2422**) a cuboid shape corresponding to the perimeter of the pallet **2502** with payload **2504**.
[0254] Method Embodiment 6. The method of Method Embodiment 5, wherein capturing (**2452**) a set of sensor data providing 3D information corresponding to the pallet storage location includes capturing a set of images of the pallet storage location using cameras of the stereoscopic camera array.
[0255] Method Embodiment 7. The method of Method Embodiment 6, wherein generating (**2454**) a 3D representation of the available storage space at the pallet placement location from captured sensor data includes generating (**2458**) a cuboid shape corresponding to the available space at the pallet placement location.
[0256] Method Embodiment 8. The method of Method Embodiment 7, wherein determining (**2462**) if there is sufficient space at the pallet placement location for successful pallet placement includes: comparing (**2464**) the cuboid shape corresponding to the perimeter of the pallet **2502** with payload **2504** to the cuboid shape corresponding to the available space at the pallet placement location.
[0257] Method Embodiment 9. The method of Method Embodiment 7, wherein determining (**2462**) if there is sufficient space at the pallet placement location for successful pallet placement includes: determining (**2470**) if the cuboid shape corresponding to the perimeter of the pallet **2502** with payload **2504** first with the cuboid shape corresponding to the available space at the pallet placement location by at least a predetermined amount (e.g., 1 foot, 6 inches or some other amount expressed as a distance) on the top, left side, right side and front portions of the cuboid shape corresponding to the perimeter of the pallet **2502** with payload **2504**.
[0258] Method Embodiment 10. The method of Method Embodiment 9, further comprising: displaying on an operator workstation **200** an image **3200** of the pallet placement location **2904** along with the cuboid **2802** corresponding to the pallet **2502** with payload **2504** with a visual indication **3104** of a predicted collision area if the pallet **2502** is inserted into the pallet placement location **2904**.
[0259] Method Embodiment 10A. The method of Method Embodiment 10 wherein the visual indication **3104** includes coloring a side of the cuboid **2802** corresponding to the pallet **2502** with payload **2504** red on the side **3104** where the collision will occur.
[0260] Method Embodiment 10B. The method of Method Embodiment 10A, wherein displaying on the operator workstation **200** the image **3200** of the pallet placement location **2904** along with the cuboid **2802** corresponding to the pallet **2502** with payload **2504** includes coloring cuboid areas (e.g., left, front top, front bottom, which are not predicted to be subject to a collision green.
[0261] Method Embodiment 10C. The method of Method Embodiment 4, wherein the robotic device **300** is a robotic forklift which supports an autonomous mode of operation and an operator controlled mode of operation; and wherein controlling (**2478**) robotic device (**300**) operation based on the determination (**2462**) of whether or not there is sufficient space at the pallet placement location for successful pallet placement includes: when it is determined there is sufficient space at the pallet storage location for storing the pallet **2502** with goods **2504** in the storage location, operating (**2490**) in an autonomous mode of robotic device operation to store the pallet **2502** with goods **2504** when there is sufficient space at the pallet placement location for successful pallet placement; and when it is determined there is not sufficient space at the pallet storage location for storing the pallet **2502** with goods **2504** in the storage location, operating (**2496**) the robotic device **300** under operator control to store the pallet **2502** with goods **2504** in said storge location or operating (**2498**) the robotic device **300** in an autonomous mode of operation to store the pallet **2502** with payload **2504** in an alternative storage location.
[0262] Method Embodiment 11. The method of Method Embodiment 3, further comprising: capturing (**2426**) (e.g., at the pallet placement location) a second set of sensor data providing 3D

information corresponding to the pallet **2502** with payload **2504** following the move (e.g., images captured by cameras of a camera system (e.g., forward looking camera system **392**) which includes multiple cameras spaced apart from one another to support stereoscopic depth and/or shape determinations which are generated from images captured by the cameras of the camera system or LIDA, RADA or SONAR sensor data providing 3D information about the pallet and payload); and generating (**2428** 3D representation of pallet with payload after move) a second 3D representation of the pallet **2502** with payload **2504** from captured sensor data; and comparing (**2436**) the first and second 3D representations of the pallet **2502** with payload **2504** to determine an amount of any shift in payload during moving of the pallet from the pickup location.

[0263] Method Embodiment 12. The method of Method Embodiment 11, further comprising: determining (**2442**) if a shift in payload which occurred while moving the pallet **2502** with payload **2504** is likely to result in failure of pallet placement (e.g., into the intended storage location).

[0264] Method Embodiment 13. The method of Method Embodiment 12, wherein determining (**2442**) if a shift in payload which occurred while moving the pallet **2502** with payload **2504** is likely to result in failure of placement of the pallet **2502** with goods **2504** (e.g., into the intended storage location).

[0265] Method Embodiment 13A. The method of Method Embodiment 13, wherein determining (**2442**) if a shift in payload which occurred while moving the pallet **2502** with payload **2504** is likely to result in failure of placement of the pallet **2502** with goods **2504** includes determining (**2444**) if a shift in goods over the edge of the pallet **2502** by a predetermined amount (e.g., 6 inches) occurred during the move from the pickup location.

[0266] Method Embodiment 14. The method of Method Embodiment 13, further comprising: in response to determining (**2442**) that a shift in payload which occurred while moving the pallet **2502** with payload **2504** is likely to result in failure of placement of the pallet **2502** with goods **2504**, transporting the pallet **2502** with payload **2504** to an inspection area, a repacking area or inspection and repacking area prior to placement of the pallet **2502** with goods **2504** in a pallet storage location.

[0267] Method Embodiment 14A. The method of Method Embodiment 14, further comprising: subjecting (**2448**) the pallet **2502** with payload **2504** to an inspection, restacking and/or repacking of goods operation to change the arrangement of goods on the pallet **2502** to avoid goods extending over the edge of the pallet **2502**.

This Set of Apparatus/System Embodiments

[0268] Apparatus Embodiment 1. A system for controlling the movement of one or more pallets, the system comprising: a robotic forklift device (**300**); one or more processors (**202**, **306**) configured to control the robotic device to: move (**2414**) to a pallet pickup location (e.g., under the control of instructions communicated by a control device such as operator workstation **200** and received by the robotic device in step **2412**); capture (**2416**) (e.g., at the pallet pickup location) a first set of sensor data providing 3D information corresponding to a pallet **2502** with payload **2504** (e.g., images captured by cameras of a camera system (e.g., forward looking camera system **392**) which includes multiple cameras spaced apart from one another to support stereoscopic depth and/or shape determinations which are generated from images captured by the cameras of the camera system or LIDA, RADA or SONAR sensor data providing 3D information about the pallet and payload); and generate (**2418** 3D representation of pallet **2502** with payload **2504** before moving or **2428** 3D representation of pallet with payload after move) a 3D representation of the pallet **2502** with payload **2504** from captured sensor data.

[0269] Apparatus Embodiment 2. The system of Apparatus Embodiment 1, wherein the one or more processors (**202**, **306**) are further configured to: control the robotic device (**300**) to capture (**2452**) (e.g., at a location in front of a storage rack into which the pallet **2502** with payload **2504** is inserted) a set of sensor data providing 3D information corresponding to a pallet storage location; (e.g., images captured by cameras of the camera system (e.g., forward looking camera system **392**) which includes multiple cameras spaced apart from one another to support stereoscopic depth and/or shape determinations which are generated from images captured by the cameras of the camera system or LIDA, RADA or SONAR sensor data providing 3D information about the pallet and payload); generate (**2454**) a 3D representation of the available storage space at the pallet placement location from captured sensor data; and determine (**2462**) if there is sufficient space at the pallet placement location for successful pallet placement (e.g., determine if the robotic device **300** is likely to succeed or will succeed with placement of the pallet **2502** with goods **2504**, e.g., autonomously without operator assistance), said determining including comparing (**2464**) the generated 3D shape corresponding to the pallet **2502** with the payload **2504** to the generated 3D shape corresponding to the available storage space at the pallet placement location.

[0270] Apparatus Embodiment 3. The system of Apparatus Embodiment 2, wherein the one or more processors (**202**, **306**) are further configured to: control (**2478**) robotic device (**300**) operation based on the determination (**2462**) of whether or not there is sufficient space at the pallet placement location for successful pallet placement.

[0271] Apparatus Embodiment 4. The system of Apparatus Embodiment 3, wherein capturing (**2416**) a first set of sensor data providing 3D information corresponding to a pallet **2502** with payload **2504** includes capturing a first set of images using cameras of a stereoscopic camera array (**392**); and wherein generating a 3D representation of the pallet **2502** with payload **2504** from captured sensor data includes processing the captured images to generate a 3D shape or model corresponding to the pallet **2502** with payload **2504**.

[0272] Apparatus Embodiment 5. The system of Apparatus Embodiment 4, wherein generating (**2418**) a 3D representation of the pallet **2502** with payload **2504** from captured sensor data includes operating the one or more processors to process the captured images to generate (**2422**) a cuboid shape corresponding to the perimeter of the pallet **2502** with payload **2504**.

[0273] Apparatus Embodiment 6. The system of Apparatus Embodiment 5, wherein the robotic device (**300**) further includes a camera array (**392**) and wherein controlling the robotic device to capture (**2452**) a set of sensor data providing 3D information corresponding to the pallet storage location includes controlling the robotic device to capture a set of images of the pallet storage location using cameras of the stereoscopic camera array.

[0274] Apparatus Embodiment 7. The system of Apparatus Embodiment 6, wherein operating the one or more processors (**202**, **306**) to generate (**2454**) a 3D representation of the available storage space at the pallet placement location from captured sensor data includes operating the one or more processors to generate (**2458**) a cuboid shape corresponding to the available space at the pallet placement location.

[0275] Apparatus Embodiment 8. The system of Apparatus Embodiment 7, wherein operating the one or more processors (**202**, **306**) to determine (**2462**) if there is sufficient space at the pallet placement location for successful pallet placement includes: operating the one or more processors (**202**, **306**) to compare (**2464**) the cuboid shape corresponding to the perimeter of the pallet **2502** with payload **2504** to the cuboid shape corresponding to the available space at the pallet placement location.

[0276] Numerous additional variations on the methods and apparatus of the present invention described above will be apparent to those skilled in the art in view of the above description of the invention. Such variations are to be considered within the scope of the invention. The methods and apparatus of the present invention may be, and in various embodiments are, implemented using a variety of wireless communications technologies such as CDMA, orthogonal frequency division multiplexing (OFDM), WiFi, and/or various other types of communications techniques which may be used to provide wireless communications links.

[0277] Some aspects and/or features are directed to a non-transitory computer readable medium embodying a set of software instructions, e.g., computer executable instructions, for controlling a computer or other device, e.g., a vehicle such as a forklift or other robotic device or an operator

workstation, to operate in accordance with the above discussed methods.
[0278] The techniques of various embodiments may be implemented using software, hardware and/or a combination of software and hardware. Various embodiments are directed to a control apparatus, e.g., controller or control system, which can be implemented using a microprocessor including a CPU, memory and one or more stored instructions for controlling a device or apparatus to implement one or more of the above described steps. Various embodiments are also directed to methods, e.g., a method of controlling a device, e.g., a forklift or other robotic device, or an operator workstation and/or performing one or more of the other operations described in the present application. Various embodiments are also directed to a non-transitory machine, e.g., computer, readable medium, e.g., ROM, RAM, CDs, hard discs, etc., which include machine readable instructions for controlling a machine to implement one or more steps of a method.
[0279] As discussed above various features of the present invention are implemented using modules and/or components. Such modules and/or components may, and in some embodiments are, implemented as software modules and/or software components. In other embodiments the modules and/or components are implemented in hardware. In still other embodiments the modules and/or components are implemented using a combination of software and hardware. In some embodiments the modules and/or components are implemented as individual circuits with each module and/or component being implemented as a circuit for performing the function to which the module and/or component corresponds. A wide variety of embodiments are contemplated including some embodiments where different modules and/or components are implemented differently, e.g., some in hardware, some in software, and some using a combination of hardware and software. It should also be noted that routines and/or subroutines, or some of the steps performed by such routines, may be implemented in dedicated hardware as opposed to software executed on a general purpose processor. Such embodiments remain within the scope of the present invention. Many of the above described methods or method steps can be implemented using machine executable instructions, such as software, included in a machine readable medium such as a memory device, e.g., RAM, floppy disk, etc. to control a machine, e.g., general purpose computer with or without additional hardware, to implement all or portions of the above described methods. Accordingly, among other things, the present invention is directed to a machine-readable medium including machine executable instructions for causing a machine, e.g., processor and associated hardware, to perform one or more of the steps of the above-described method(s).
[0280] The techniques of the present invention may be implemented using software, hardware and/or a combination of software and hardware. The present invention is directed to apparatus, e.g., a vehicle which implements one or more of the steps of the present invention. The present invention is also directed to machine readable medium, e.g., ROM, RAM, CDs, hard discs, etc., which include machine readable instructions for controlling a machine to implement one or more steps in accordance with the present invention.
[0281] Numerous additional variations on the methods and apparatus of the various embodiments described above will be apparent to those skilled in the art in view of the above description. Such variations are to be considered within the scope.

Claims

1. A method of controlling a robotic material handling device comprising: controlling the robotic material handling device to perform a first pallet pickup operation to pick up a pallet; and determining from the output of a pallet position sensor on the robotic material handling device if the pallet is nested on forks of the robotic material handling device.
2. The method of claim 1, further comprising: controlling the robotic material handling device, in response to determining from the output of the pallet position sensor that the pallet is not nested on the forks, to move the pallet closer to the robotic material handling device while the pallet is in the

air on the forks due to the first pallet pickup; controlling the robotic material handling device to deposit the pallet; and controlling the robotic material handling device to perform a second pallet pickup operation to pick up the pallet for a second time.

3. The method of claim 2, further comprising: determining, after the second pallet pickup operation, from the output of the pallet position sensor on the robotic material handling device if the pallet is nested on forks of the robotic material handling device.

4. The method of claim 3, further comprising: in response to determining, after the second pallet pickup operation, that the pallet is nested on forks, transporting the pallet to a pallet destination.

5. The method of claim 1, wherein said pallet position sensor is mounted at a fixed position on a fork backrest that will move with the forks as the forks are extended or retracted.

6. The method of claim 5, wherein said pallet position sensor is a non-contact distance measuring device.

7. The method of claim 6, wherein said pallet position sensor is mounted on a member of the fork backrest positioned at a location which is between the forks.

8. The method of claim 7 wherein the pallet position sensor is mounted above a forward looking camera array which has a field of view including the pallet forks.

9. The method of claim 4, further comprising: receiving an instruction to move the pallet to the pallet destination; determining the location of the pallet to be moved; determining a safe forklift placement location from which to pick the pallet to be moved; controlling the robotic material handling device to move to the determined forklift placement location prior to proceeding with lifting of the pallet.

10. The method of claim 1, wherein the robotic material handling device is a robotic forklift capable of operating in a fully autonomous mode in which the processor controls the robotic device to perform the recited steps.

11. A robotic material handling device comprising: a fork assembly including a backrest, a first fork and a second fork; a pallet position sensor; and a processor coupled to the pallet position sensor configured to control the robotic material handling device to: perform a first pallet pickup operation to pick up a pallet; and determine from the output of a pallet position sensor on the robotic material handling device if the pallet is nested on forks of the robotic material handling device.

12. The robotic material handling device of claim 11, wherein the processor is further configured to control the robotic material handling device to: move the pallet closer to the robotic material handling device while the pallet is in the air on the forks due to the first pallet pickup, in response to determining from the output of the pallet position sensor that the pallet is not nested on the forks; deposit the pallet; and perform a second pallet pickup operation to pick up the pallet for a second time.

13. The robotic material handling device of claim 12, further comprising: determining after the second pallet pickup operation, from the output of the pallet position sensor on the robotic material handling device if the pallet is nested on forks of the robotic material handling device.

14. The robotic material handling device of claim 13, further comprising: in response to determining, after the second pallet pickup operation (**1020**), that the pallet is nested on forks, transporting the pallet to a pallet destination.

15. The robotic material handling device of claim 11, wherein said pallet position sensor is mounted at a fixed position on a fork backrest that will move with the forks as the forks are extended or retracted.

16. The robotic material handling device of claim 15, wherein said pallet position sensor is a non-contact distance measuring device.

17. The robotic material handling device of claim 16, wherein said pallet position sensor is mounted on a member of the fork backrest positioned at a location which is between the forks.

18. The robotic material handling device of claim 17 wherein the pallet position sensor is mounted above a forward looking camera array which has a field of view including the pallet forks.

19. The robotic material handling device of claim 14, further comprising: receiving an instruction to move the pallet to the pallet destination; determining the location of the pallet to be moved; determining a safe forklift placement location from which to pick the pallet to be moved; controlling the robotic material handling device to move to the determined forklift placement location prior to proceeding with lifting of the pallet.

20. A non-transitory computer readable medium which, when executed by a processor in a robotic material handling device controls the robotic material handling device to: perform a first pallet pickup operation to pick up a pallet; and determine from the output of a pallet position sensor on the robotic material handling device if the pallet is nested on forks of the robotic material handling device.